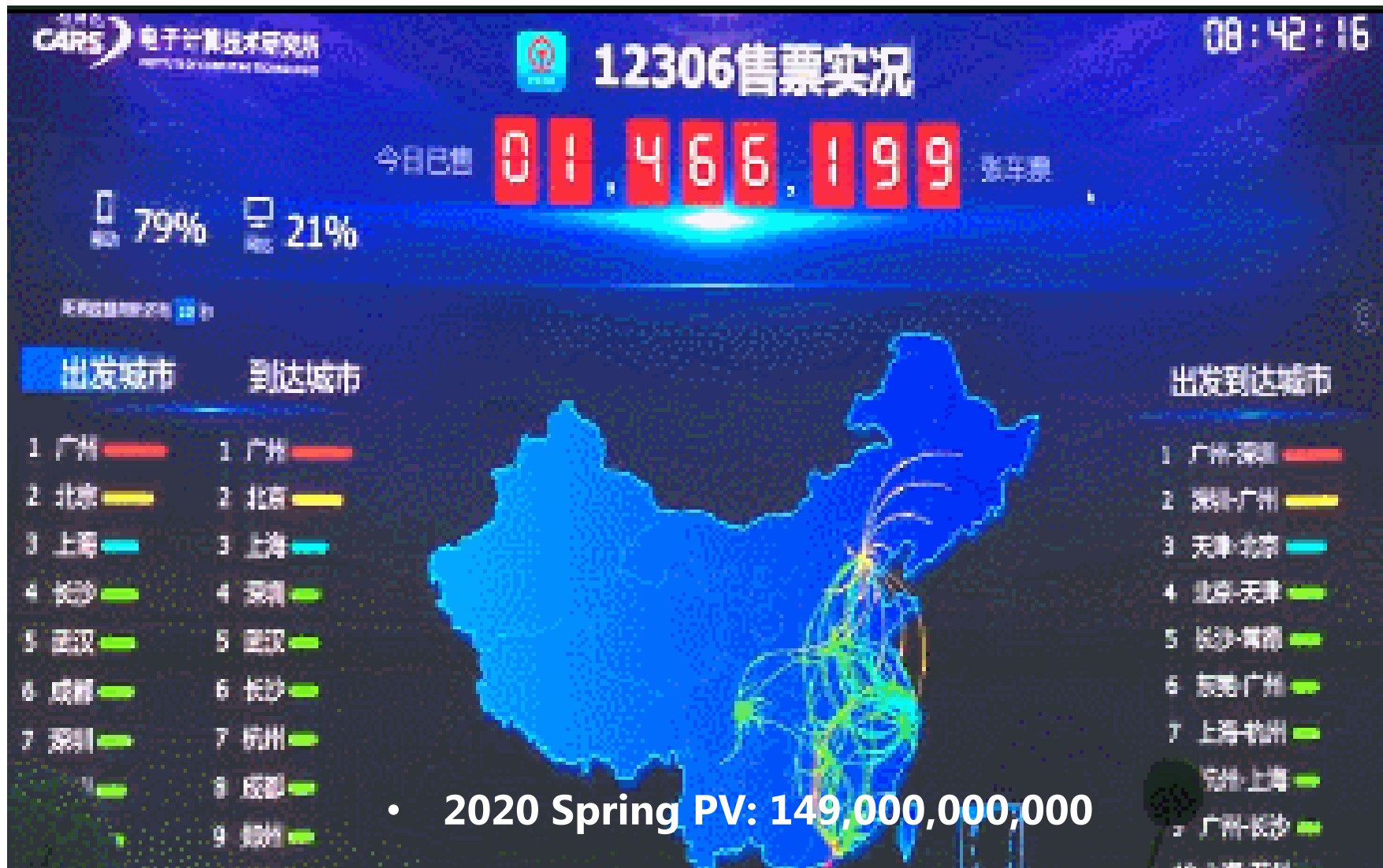


Technologies for E-Commerce

CAN302

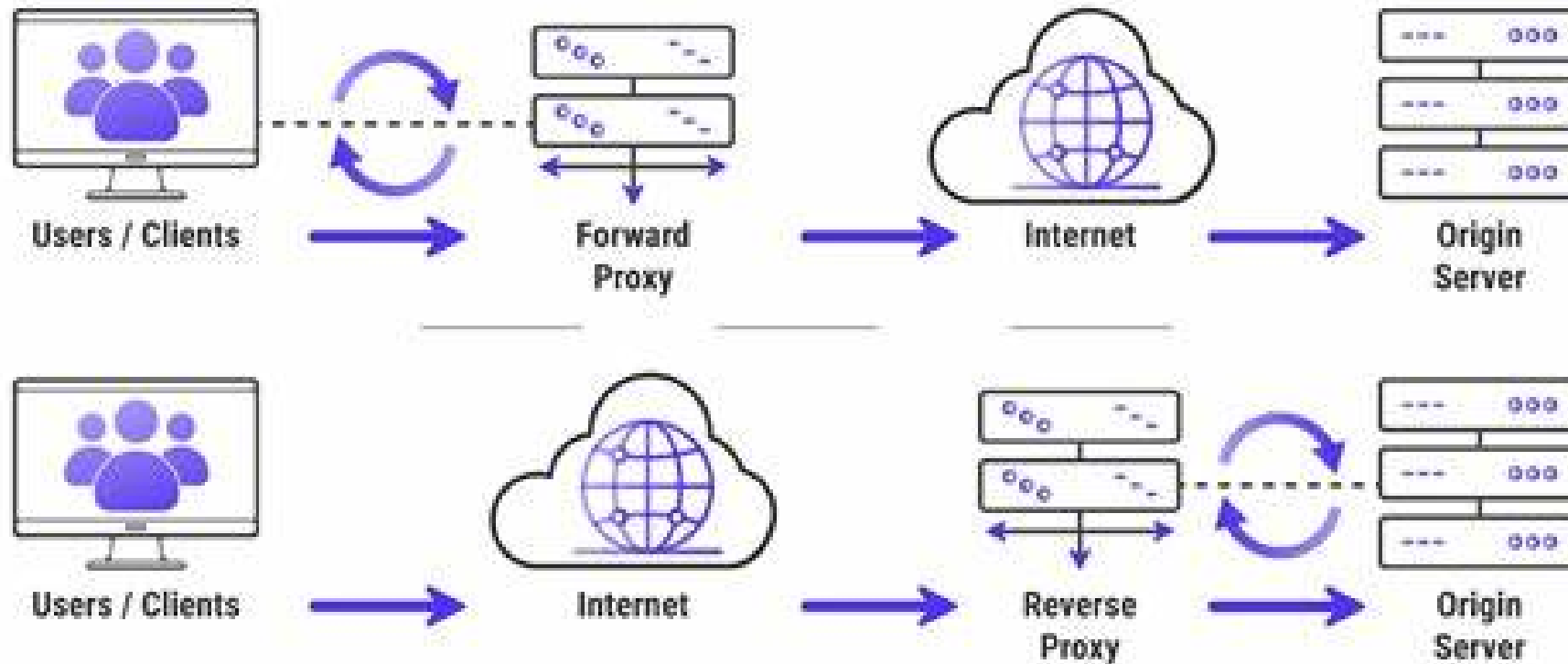
**Department of Communications and Networking
Xi'an Jiaotong-Liverpool University (XJTLU)**

Lecture 11 – Billion level traffic architecture

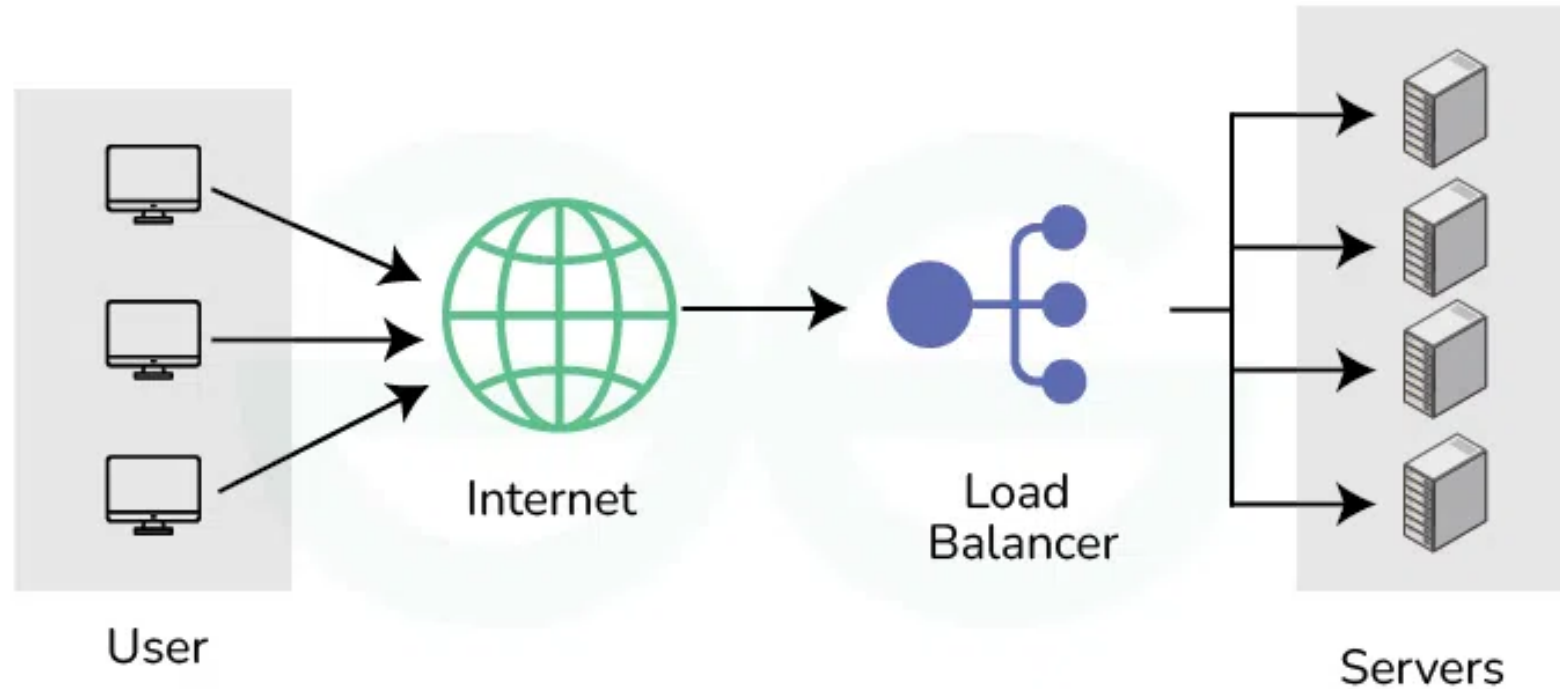


Proxy

Forward Proxy vs Reverse Proxy



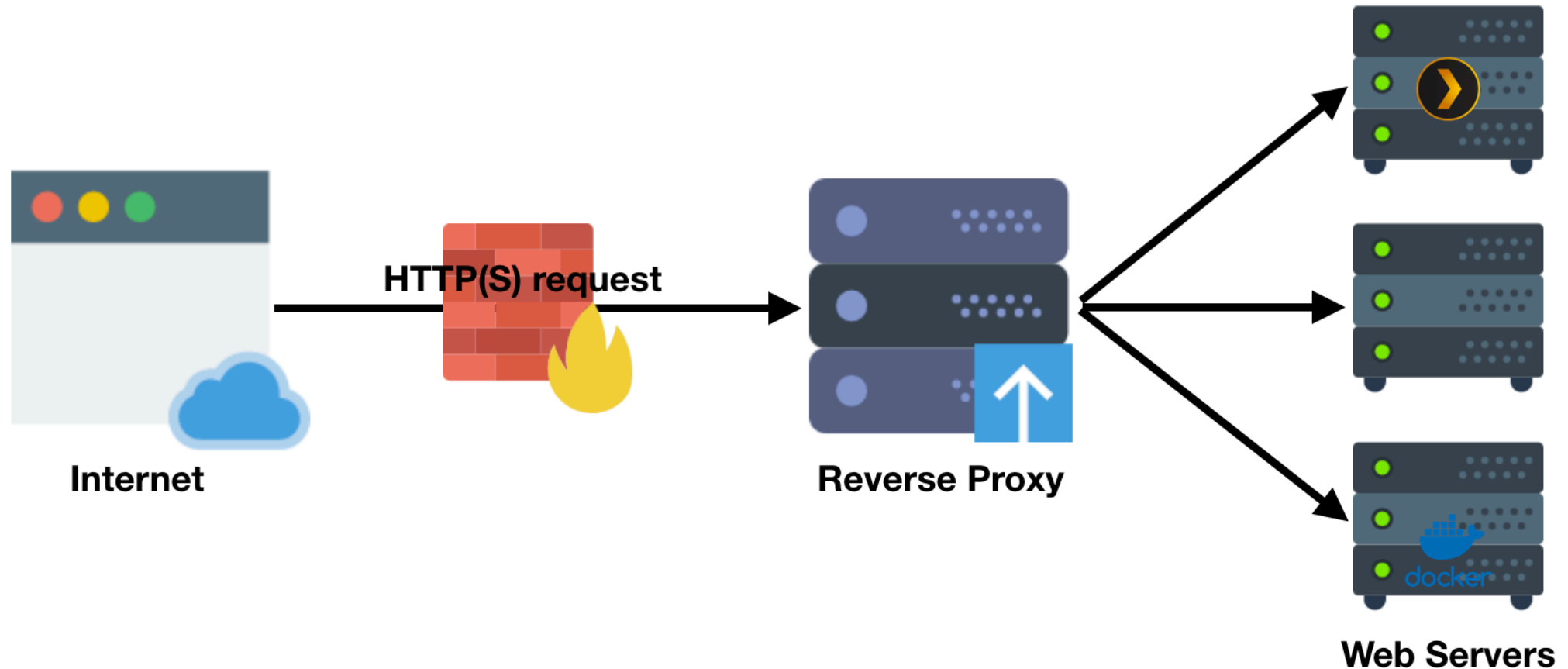
Load balance



- Load balance is similar to reverse proxy but have different emphasizes.
- A load balancer is a networking device or software application that distributes and balances the incoming traffic among the servers to provide high availability, efficient utilization of servers, and high performance.

<https://www.geeksforgeeks.org/reverse-proxy-vs-load-balancer/>

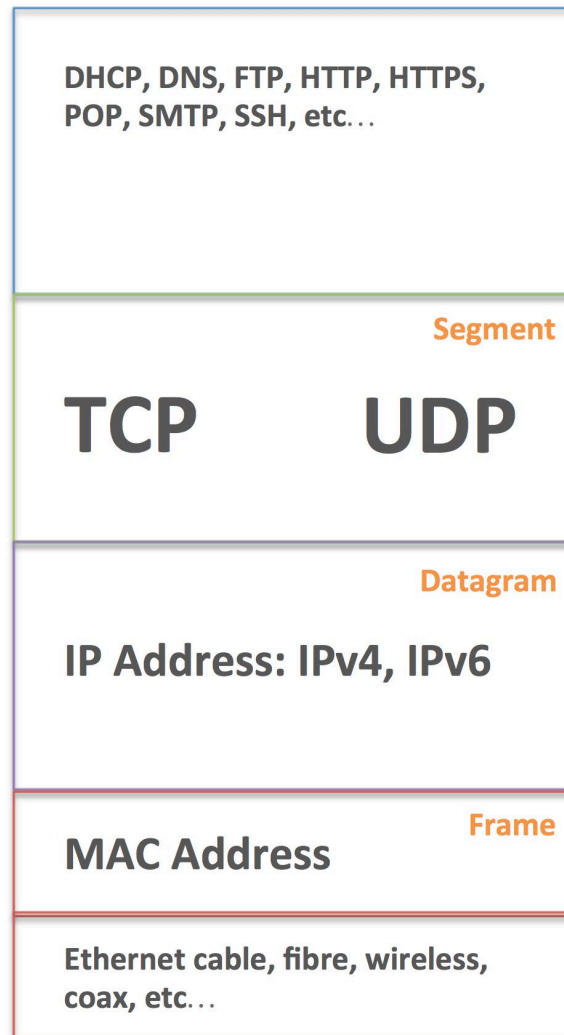
Web Proxy – 4 or 7 layer



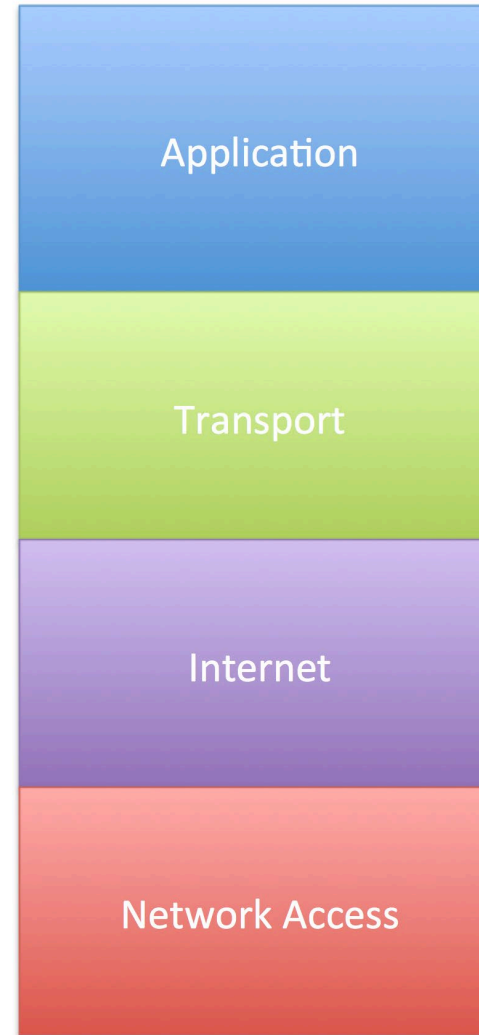
- We need a **reverse proxy** to let lots of web servers work together!

OSI and TCP/IP model

The OSI Model



The TCP/IP Model



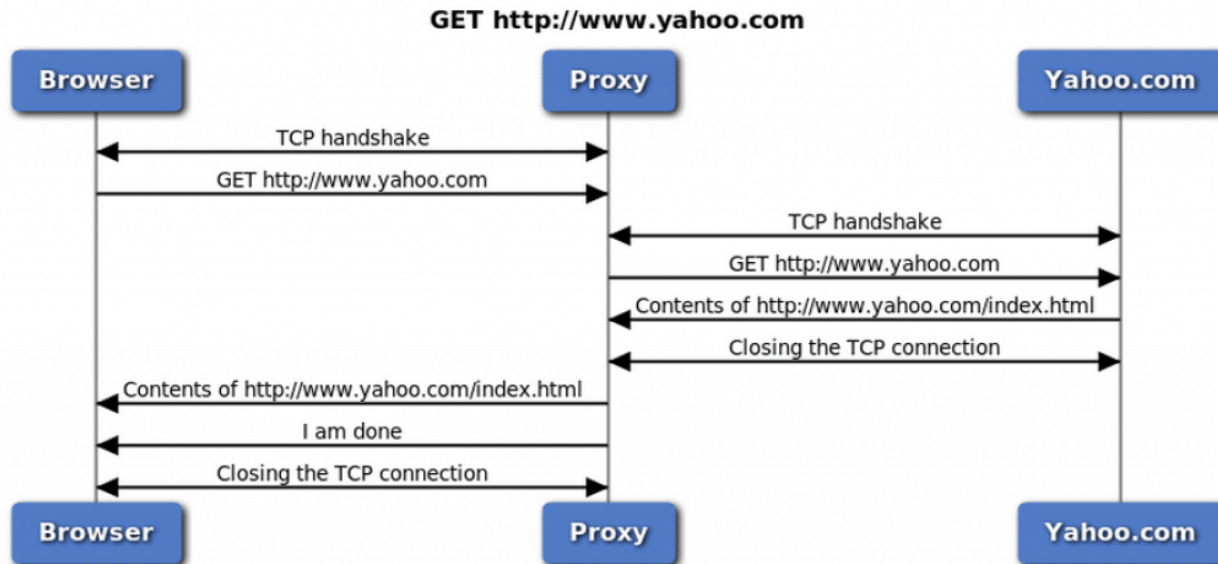
Software

Operating system

Network devices

Physical links

HTTP proxy – layer 7



Hypertext Transfer Protocol

```
> GET /portal.php HTTP/1.0\r\n
Host: [REDACTED]\r\n
X-Real-IP: 123.57.117.153\r\n
X-Forwarded-For: 220.181.108.88, 123.57.117.153\r\n
Connection: close\r\n
User-Agent: Mozilla/5.0 (compatible; Baiduspider/2.0; +ht
X-True-IP: 220.181.108.88\r\n
Web-Server-Type: nginx\r\n
WL-Proxy-Client-IP: 220.181.108.88\r\n
X-Forwarded-Proto: http\r\n
X-Forwarded-Cluster: waf,\r\n
EagleEye-TraceId: 7b39758515571054298494117ee8a6\r\n
Accept-Language: zh-cn,zh-tw\r\n
Accept-Encoding: gzip\r\n
```

- Proxy talks with both client and back server with **HTTP** request/response
- Http proxy server itself must be a HTTP server! It is also called layer 7 proxy

Apache has the proxy function

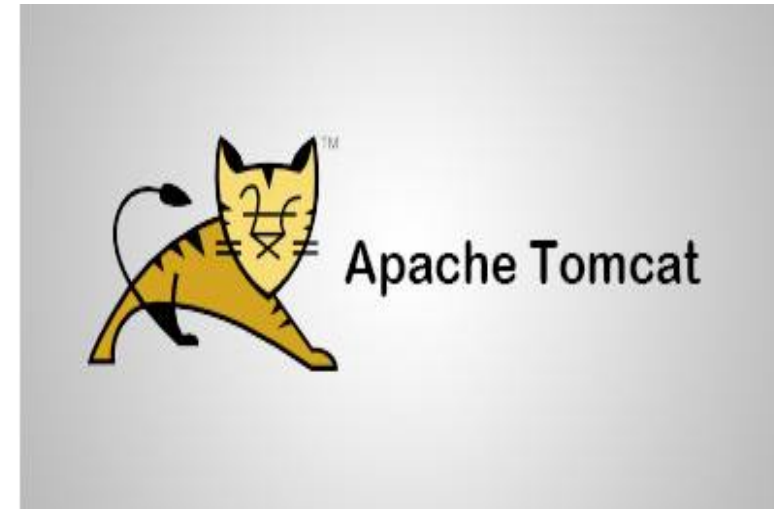


Apache

```
LoadModule unique_id_module modules/mod_unique_id.so
LoadModule setenvif_module modules/mod_setenvif.so
LoadModule version_module modules/mod_version.so
LoadModule remoteip_module modules/mod_remoteip.so
LoadModule proxy_module modules/mod_proxy.so
LoadModule proxy_connect_module modules/mod_proxy_connect.so
LoadModule proxy_ftp_module modules/mod_proxy_ftp.so
LoadModule proxy_http_module modules/mod_proxy_http.so
LoadModule proxy_fcgi_module modules/mod_proxy_fcgi.so
LoadModule proxy_scgi_module modules/mod_proxy_scgi.so
LoadModule proxy_ajp_module modules/mod_proxy_ajp.so
LoadModule proxy_balancer_module modules/mod_proxy_balancer.so
LoadModule proxy_express_module modules/mod_proxy_express.so
LoadModule session_module modules/mod_session.so
LoadModule session_cookie_module modules/mod_session_cookie.so
```

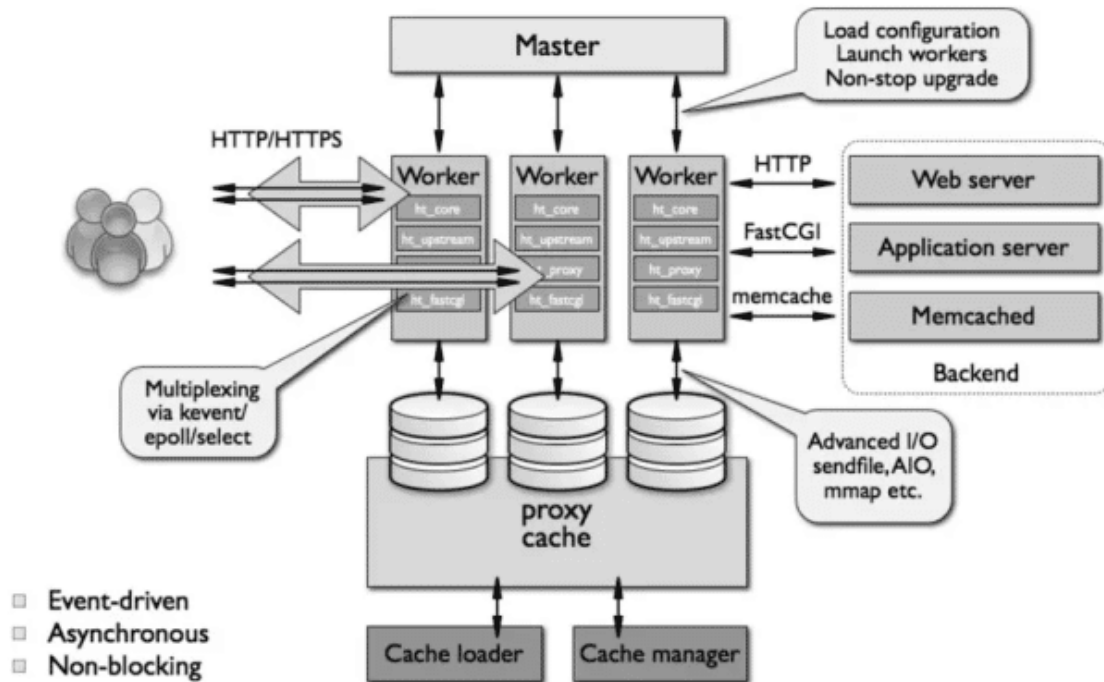
```
ProxyPass /ServerTest http://localhost:8080/
ProxyPassReverse /ServerTest http://localhost:8080/
```

```
httpd.conf
Q htdoc
# DocumentRoot: The directory out of which you will serve your
# documents. By default, all requests are taken from this directory, but
# symbolic links and aliases may be used to point to other locations.
#
DocumentRoot "/Applications/XAMPP/xamppfiles/htdocs"
<Directory "/Applications/XAMPP/xamppfiles/htdocs">
#
# Possible values for the Options directive are "None", "All",
# or any combination of:
#   Indexes Includes FollowSymLinks SymLinksifOwnerMatch ExecCGI MultiViews
#
# Note that "MultiViews" must be named *explicitly* --- "Options All"
# doesn't give it to you.
#
# The Options directive is both complicated and important. Please see
```



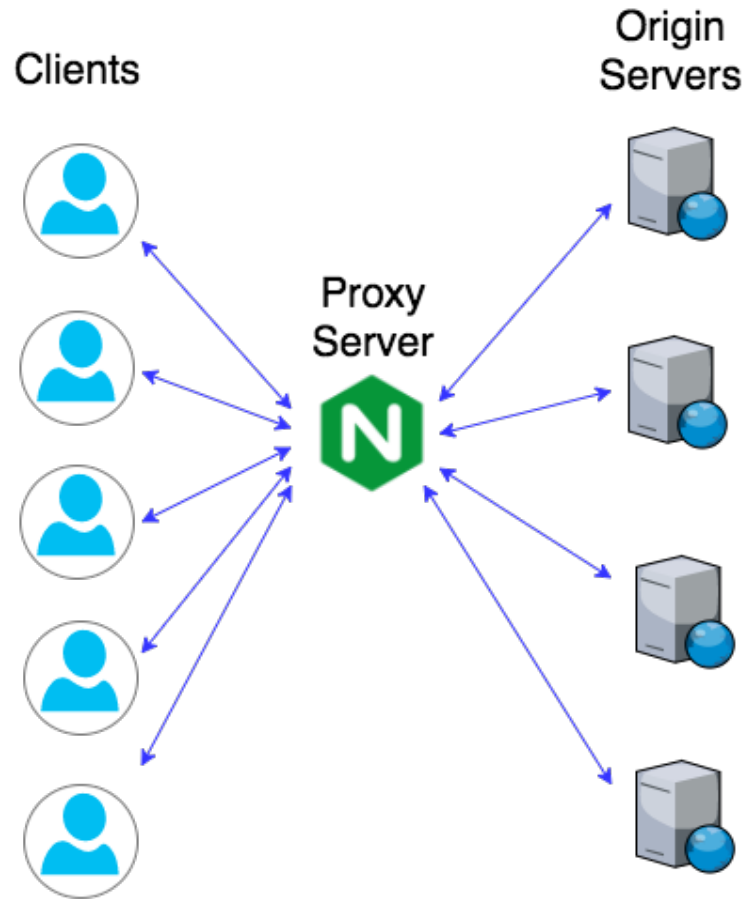
- Proxy rules are based on URL matching

Nginx - the King of soft HTTP Proxy



- Nginx is most efficient proxy server (software version).
- Nginx can server static contents.
- Nginx use multi-thread to gain the high performance.

Proxy the webserver^s



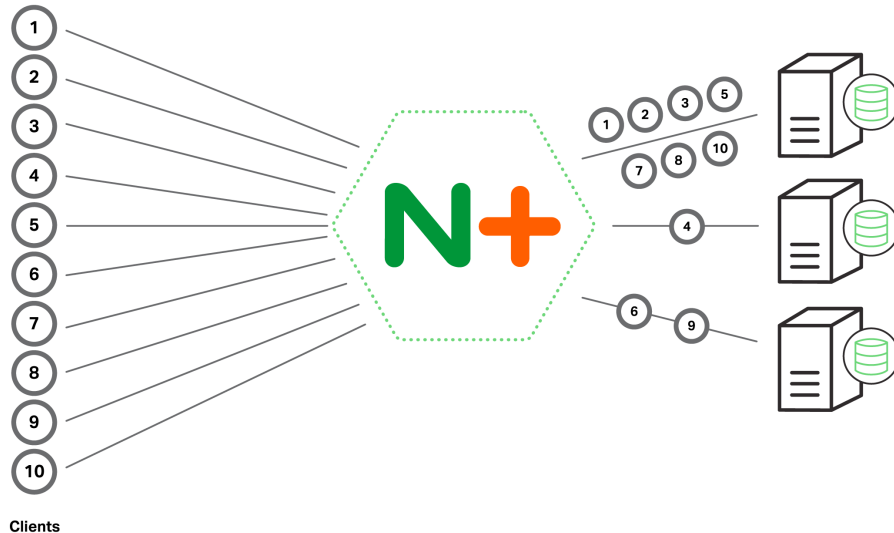
```
location /some/path/ {  
    proxy_pass http://www.example.com/link/;  
}
```

- The web should only publish one domain name to end user.
- Nginx is a strong web server.
- Nginx can server the static contents.
- Nginx set the proxy rules based on URL **path**.
- There are 4 (can be any number) servers for dynamic contents proxying together by nginx.

<http://sawers.com/blog/reverse-proxying-with-nginx/>

<https://docs.nginx.com/nginx/admin-guide/web-server/reverse-proxy/>

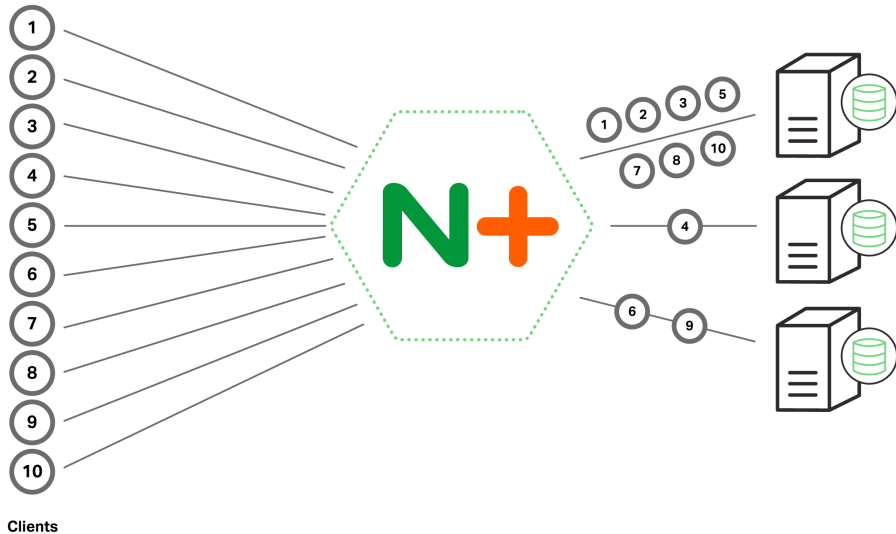
Balancing techniques



```
1 upstream backend {  
2     server web1 weight=6;  
3     server web2 weight=3;  
4     server web3;  
5 }
```

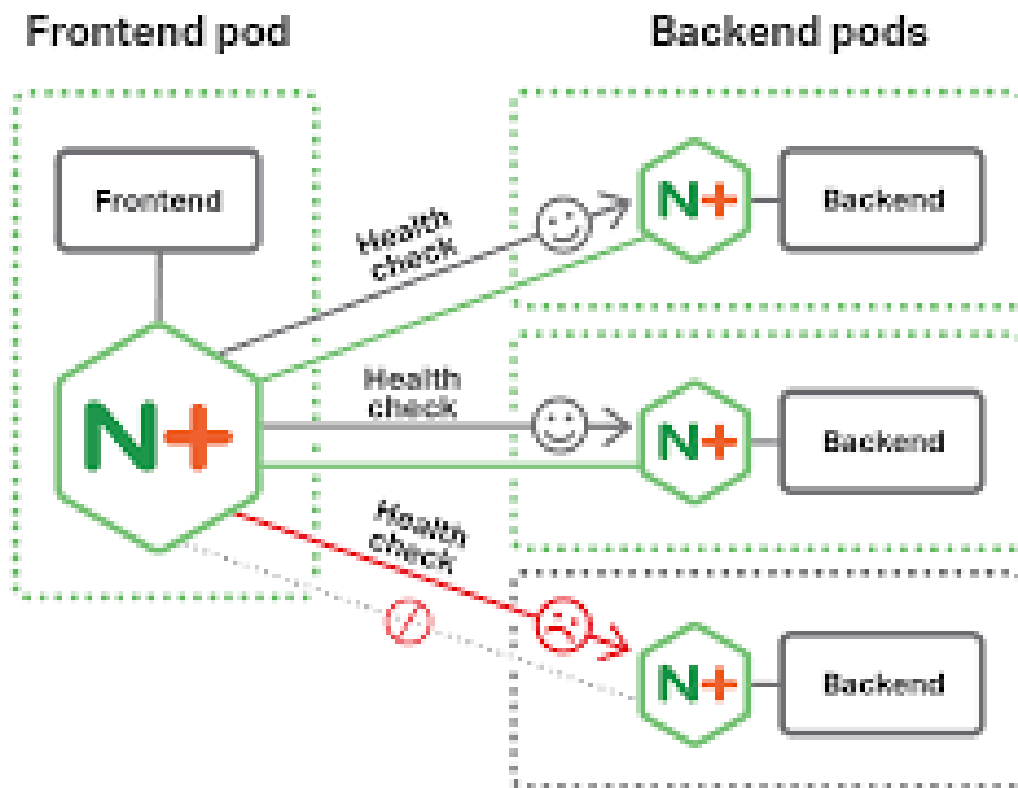
- Setting weight
 - Different server many have different capability
 - A/B test

Balancing techniques



- **Round Robin (the default method):** the load balancer runs through the list of upstream servers in sequence
- **HASH:** calculates a hash that is based on the combination of text and [NGINX variables](#) you specify
- **IP HASH:** the hash is based on the client's IP address
- **Least connections:** compares the current number of active connections and sends the request to the server with the fewest connections.
- **Least time:** combines two metrics for each server – the current number of active connections and a weighted average response time for past requests

Health check



```
location / {  
    proxy_pass http://backend;  
    health_check interval=10 fails=3 passes=2;  
}
```

```
location / {  
    proxy_pass http://backend;  
    health_check uri=/some/path;  
}
```

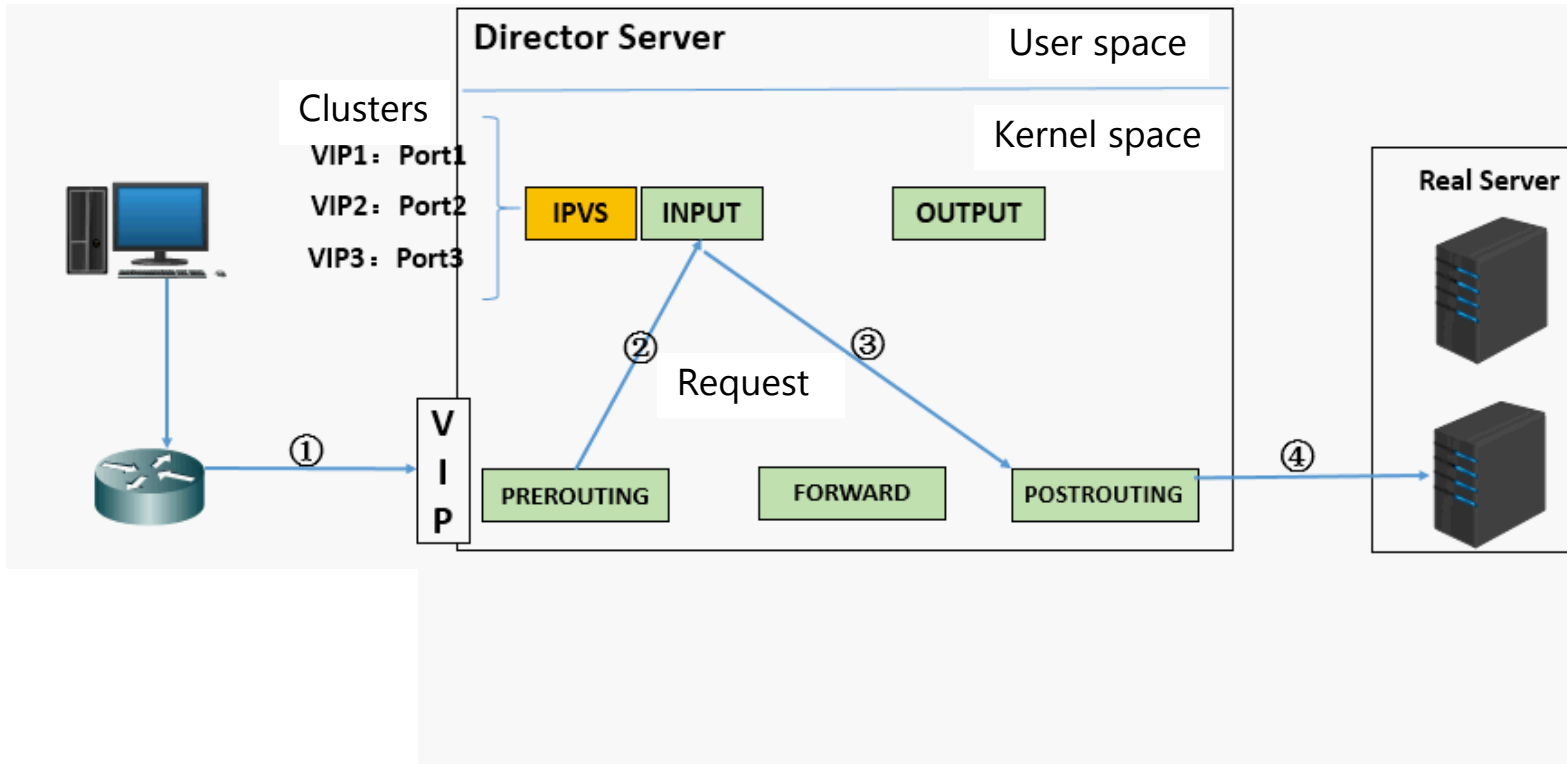
- Use health_check to make sure all backend can work properly.

Up limitation for Nginx



- Up limitation depends on the many factors
- Some report say it can support **30,000** con-concurrency with **sufficient** hardware resources
- Nginx processes the request/response in HTTP Level
- To improve the load balance further, process the flow in network layer

TCP/UDP proxy – layer 4



- LVS (Linux virtual server) is a kernel service of Linux system
- LVS share the VIP (virtual IP) with real servers
- LVS re-direct the request (TCP or UDP) packages to real servers
- Proxy rules based on **ip+ports**
- Bandwidth of network card would be the bottle neck

<http://www.linuxvirtualserver.org/>

<https://www.redhat.com/en/topics/linux/what-is-the-linux-kernel>

Network and internet layer: MAC and IP

MAC address vs IP address

MAC address

- 48 bit address
- Works at OSI layer 2 (link layer)
- Physical address
- Fixed, assigned by manufacturer

00:0C:F5:09:56:E8

IP address

- 32 bit address
- Works at OSI layer 3 (network layer)
- Logical address
- Can change depending on the network environment

150.60.122.98

```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\Administrator>ipconfig /all

Windows IP Configuration

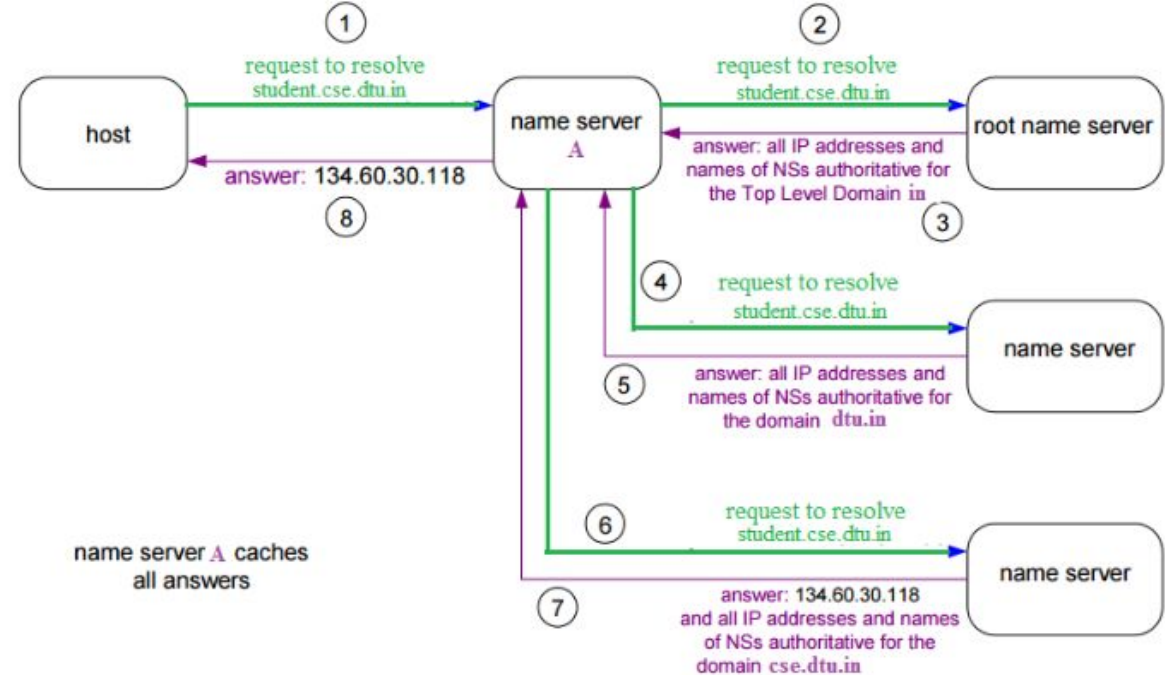
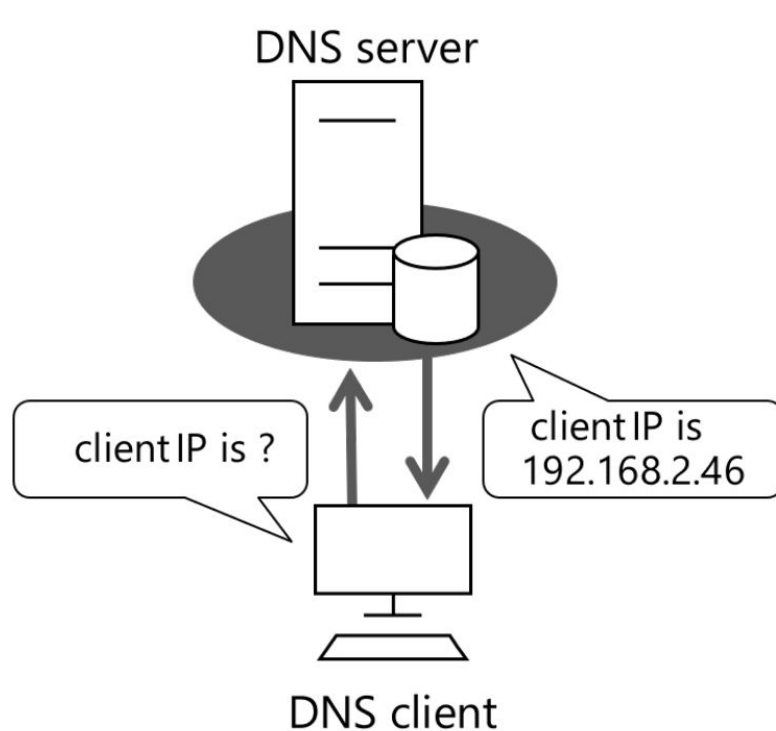
    Host Name . . . . . : winxpp
    Primary Dns Suffix . . . . . : 
    Node Type . . . . . : Unknown
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . : 
    Description . . . . . : VMware Accelerated AMD PCNet Adapter
    Physical Address. . . . . : 00-0C-29-92-3B-54
    Dhcp Enabled. . . . . : No
    IP Address. . . . . : 192.168.1.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
    DNS Servers . . . . . : 192.168.1.1

C:\Documents and Settings\Administrator>
```

Domain name & DNS



- Do you prefer remember 221.229.203.214 or www.taobao.com?
- Maybe search "Taobao" in www.baidu.com

IPv4 and IPv6

	Internet Protocol version 4 (IPv4)	Internet Protocol version 6 (IPv6)
Deployed	1981	1999
Address Size	32-bit number	128-bit number
Address Format	Dotted Decimal Notation: 192.149.252.76	Hexadecimal Notation: 3FFE:F200:0234:AB00: 0123:4567:8901:ABCD
Prefix Notation	192.149.0.0/24	3FFE:F200:0234::/48
Number of Addresses	$2^{32} = \sim 4,294,967,296$	$2^{128} = \sim 340,282,366,920,938,463,463,374,607,431,768,211,456$

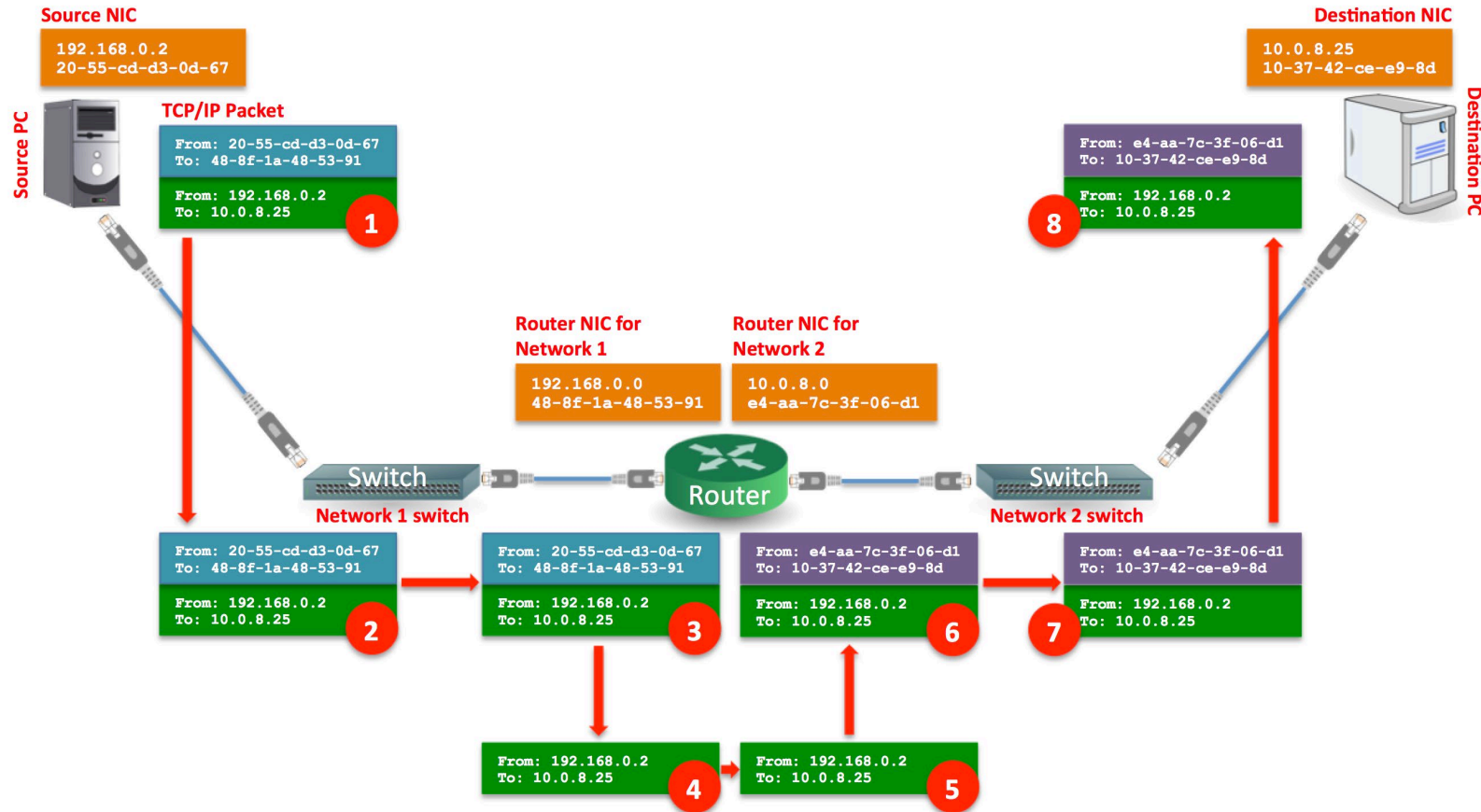
Network and internet layer: MAC and IP

Network switches



Network and internet layer: MAC and IP

The journey of a TCP/IP packet across networks

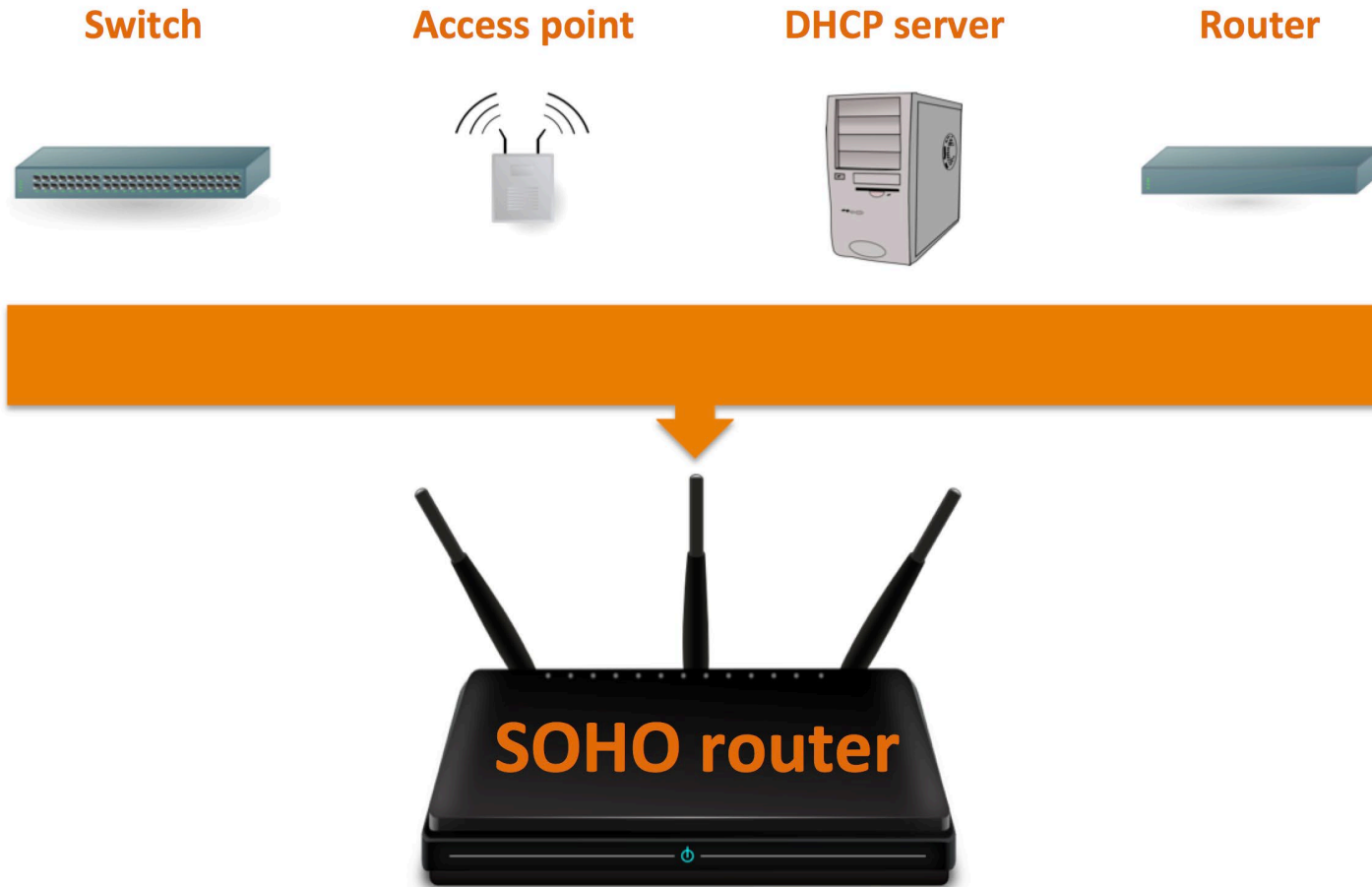


**Switch/Hub –
Create networks**

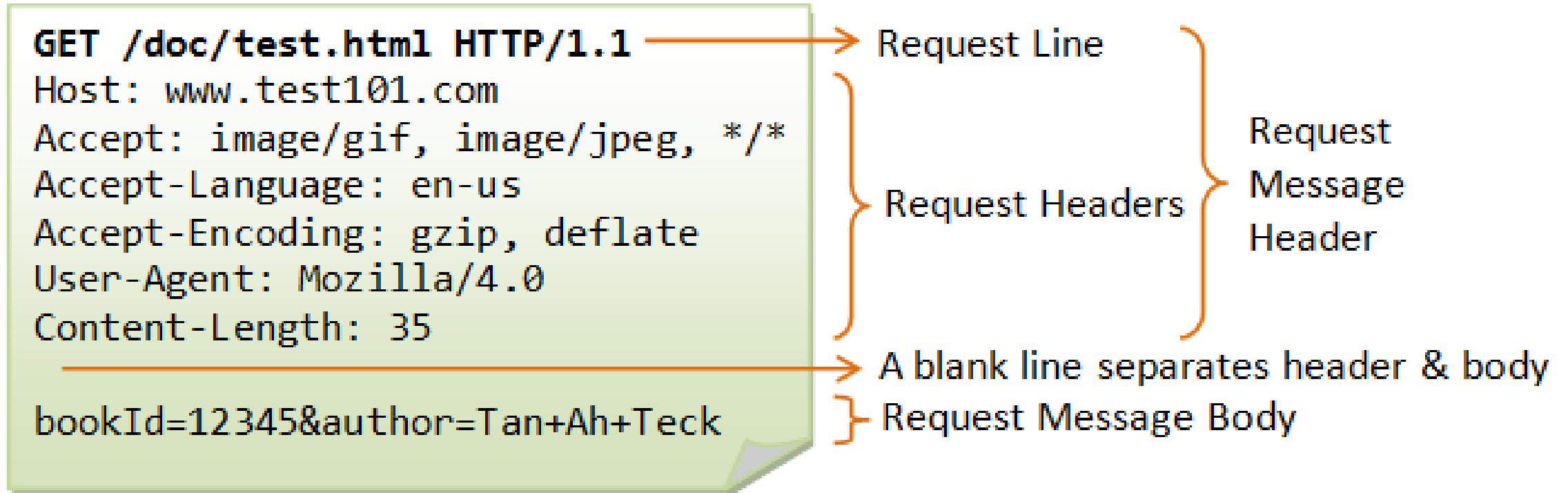
**Router –
Connect networks**

Soho router: not only a router

A SOHO router derives from the merging of traditionally distinct network hardware in a single device

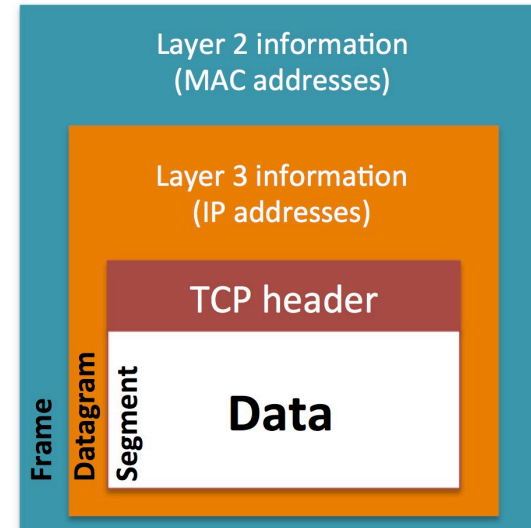


HTTP and TCP

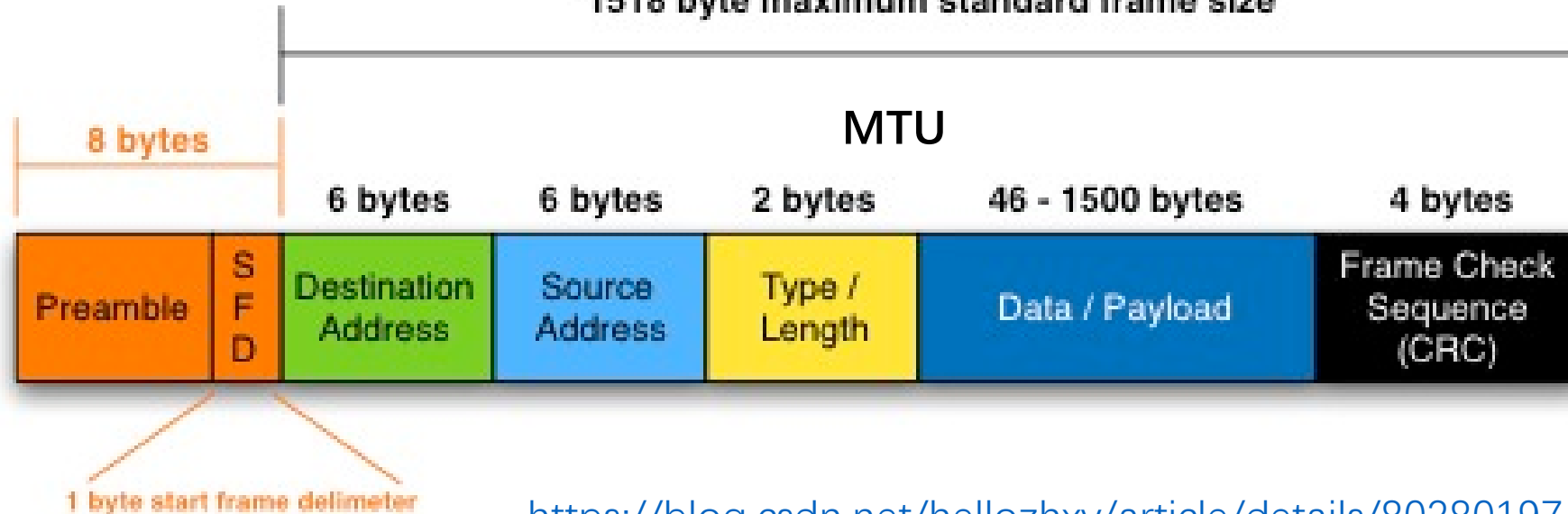


- Http request/response will be transferred in TCP packages

Russia Matryona: encapsulate the data

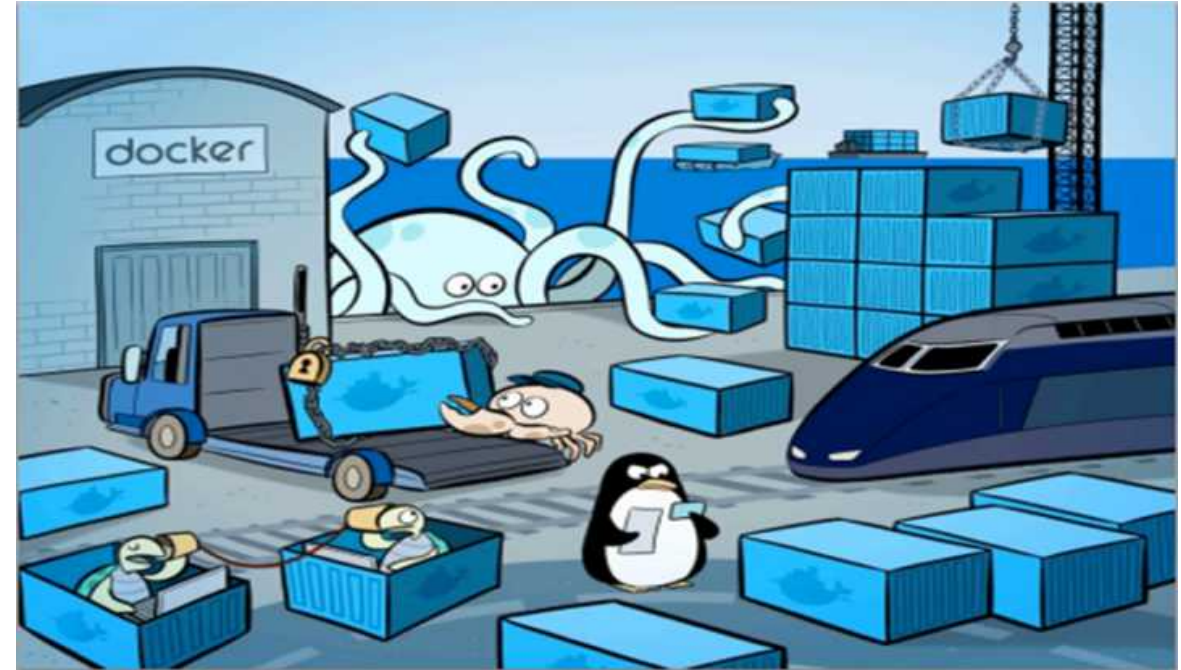
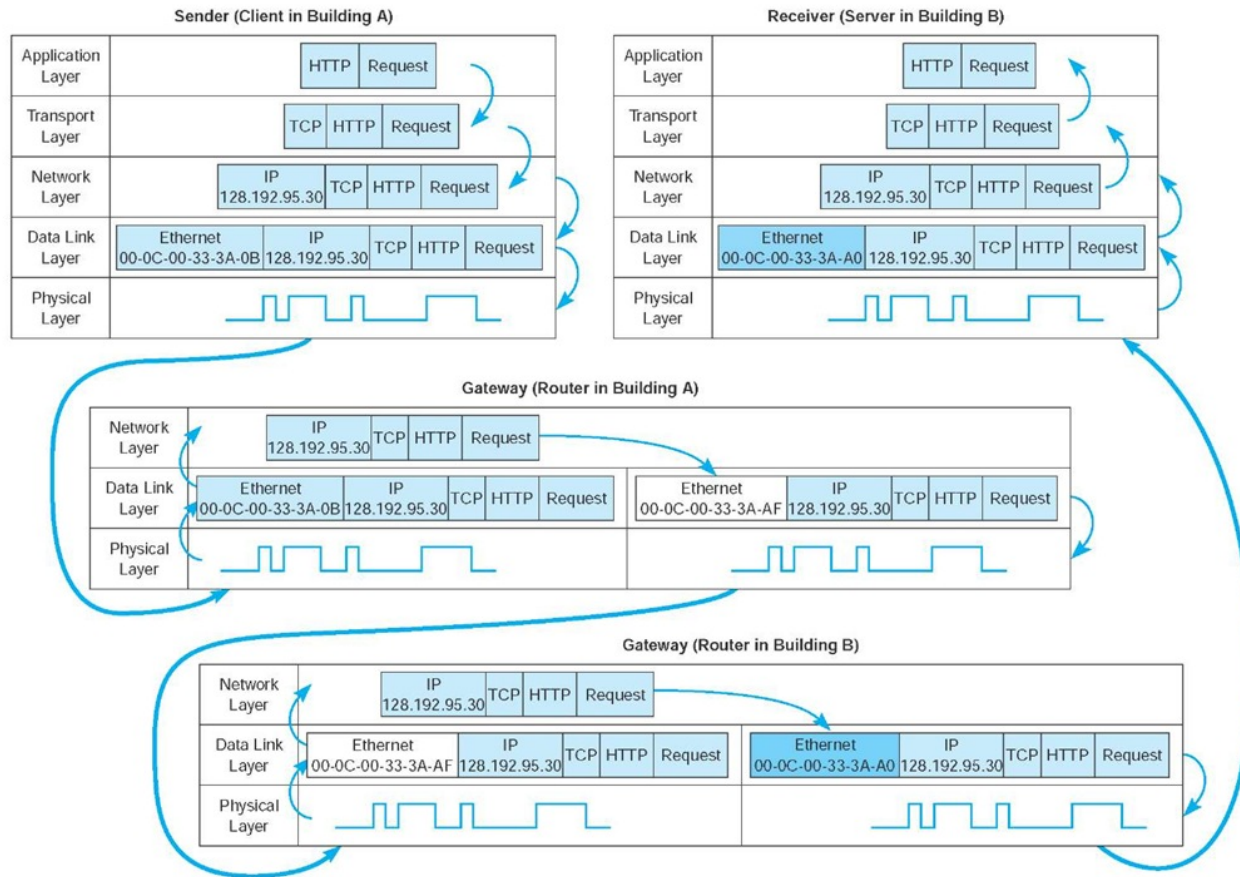


1518 byte maximum standard frame size



<https://blog.csdn.net/hellozhxy/article/details/80280197>

HTTP and TCP



- Every layer has a translation, it consumes resources
- One http request/response may split to several TCP packages

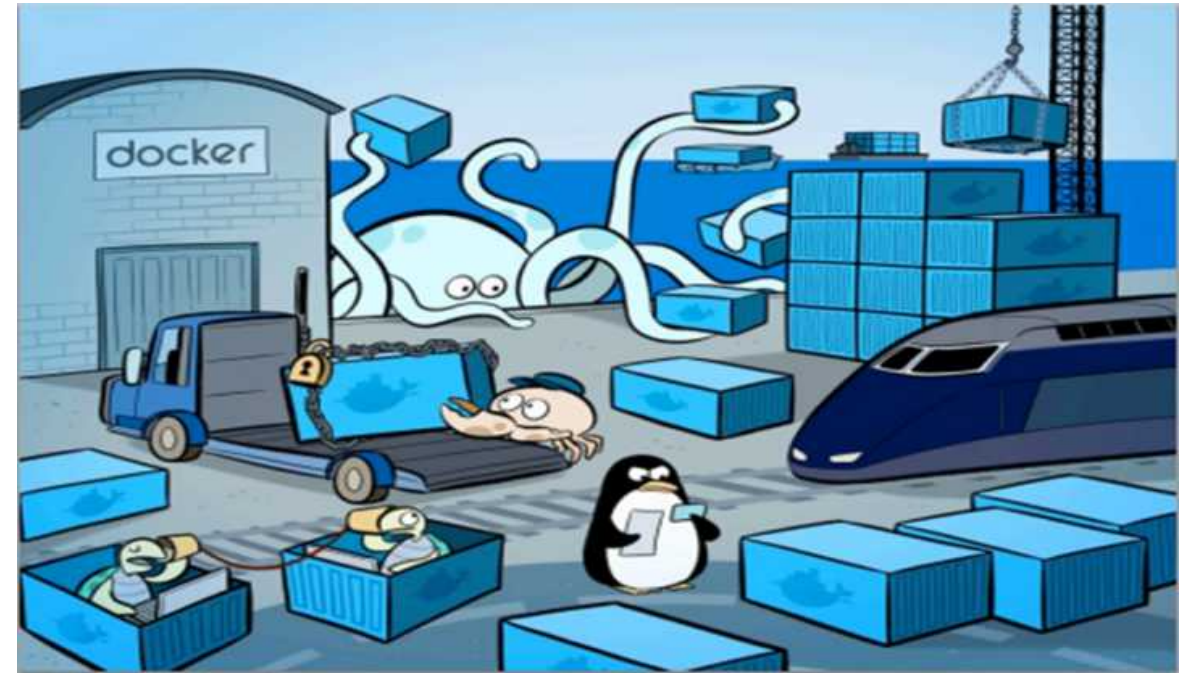
HTTP and TCP

The image shows a Wireshark packet capture of a network session. The top pane displays a list of packets, with packet 349 selected. The middle pane shows the details of this packet, which is a DNS Standard query response. The bottom pane shows the raw packet data in hexadecimal and ASCII. The packet is a DNS response from 192.168.0.1 to 192.168.0.21, containing the IP address 65.321733 for the domain cdn-0.nflximg.com.

No.	Time	Source	Destination	Protocol	Length	Info
343	65.142415	192.168.0.21	174.129.249.228	TCP	66	40555 → 80 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=491519346 TSecr=551811827
344	65.142715	192.168.0.21	174.129.249.228	HTTP	253	GET /clients/netflix/flash/application.swf?flash_version=flash_lite_2.1&v=1.5&n...
345	65.230738	174.129.249.228	192.168.0.21	TCP	66	80 → 40555 [ACK] Seq=1 Ack=188 Win=6864 Len=0 TSval=551811850 TSecr=491519347
346	65.240742	174.129.249.228	192.168.0.21	HTTP	828	HTTP/1.1 302 Moved Temporarily
347	65.241592	192.168.0.21	174.129.249.228	TCP	66	40555 → 80 [ACK] Seq=188 Ack=763 Win=7424 Len=0 TSval=491519446 TSecr=551811852
348	65.242532	192.168.0.21	192.168.0.1	DNS	77	Standard query 0x2188 A cdn-0.nflximg.com
349	65.276870	192.168.0.1	192.168.0.21	DNS	489	Standard query response 0x2188 A cdn-0.nflximg.com CNAME images.netflix.com.edge...
350	65.277992	192.168.0.21	63.80.242.48	TCP	74	37063 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=491519482 TSecr=...
351	65.297757	63.80.242.48	192.168.0.21	TCP	74	80 → 37063 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM=1 TSval=3295...
353	65.298687	192.168.0.21	63.80.242.48	HTTP	153	GET /us/nrd/clients/flash/814540.bun HTTP/1.1
354	65.318730	63.80.242.48	192.168.0.21	TCP	66	80 → 37063 [ACK] Seq=1 Ack=88 Win=5792 Len=0 TSval=3295534151 TSecr=491519503
355	65.321733	63.80.242.48	192.168.0.21	TCP	1514	[TCP segment of a reassembled PDU]

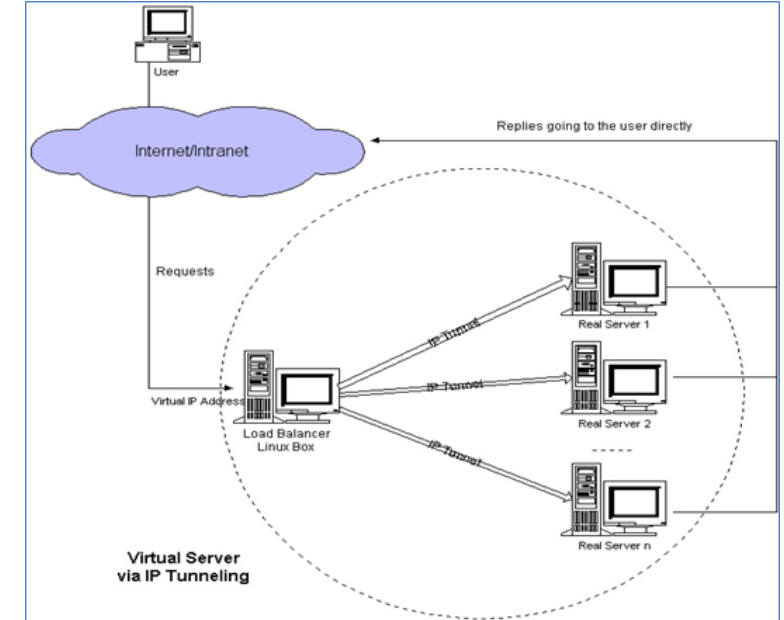
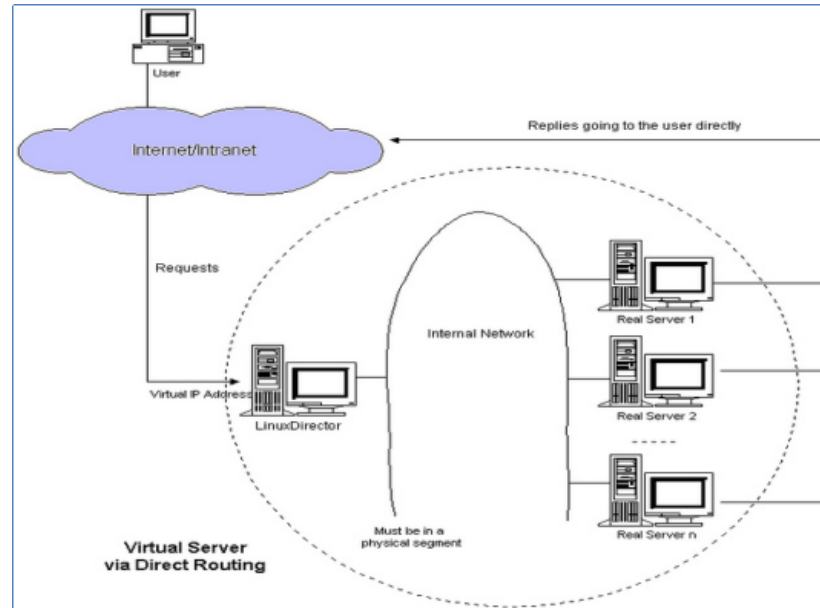
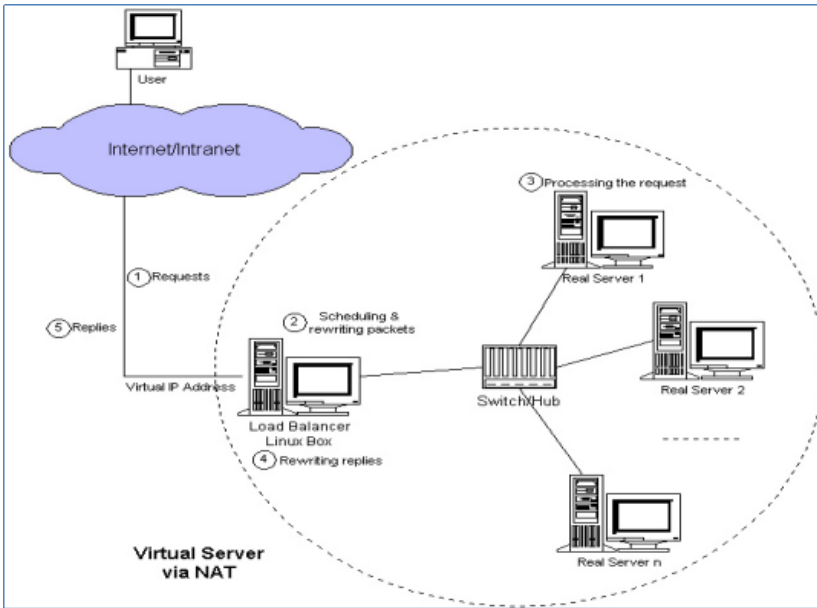
Frame 349: 489 bytes on wire (3912 bits), 489 bytes captured (3912 bits)
> Ethernet II, Src: Globalsc_00:3b:0a (f8:ad:4e:00:3b:0a), Dst: Vizio_14:8a:e1 (00:19:9d:14:8a:e1)
> Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.21
> User Datagram Protocol, Src Port: 53 (53), Dst Port: 34036 (34036)
▼ Domain Name System (response)
[Request In: 348]
[Time: 0.034338000 seconds]
Transaction ID: 0x2188
> Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 4
Authority RRs: 9
Additional RRs: 9
▼ Queries
> cdn-0.nflximg.com: type A, class IN
> Answers
> Authoritative nameservers

```
0020 00 15 00 35 84 f4 01 c7 83 3f 21 88 81 80 00 01 ...5.....?.....  
0030 00 04 00 09 00 09 05 63 64 6e 2d 30 07 6e 66 6c .....c dn-0.nfl  
0040 78 69 6d 67 03 63 6f 6d 00 00 01 00 01 c0 0c 00 ximg.com .....  
0050 05 00 01 00 00 05 29 00 22 06 69 6d 61 67 65 73 .....).images  
0060 07 6e 65 74 66 6c 69 78 03 63 6f 6d 09 65 64 67 .netflix .com.edg  
0070 65 73 75 69 74 65 03 6e 65 74 00 c0 2f 00 05 00 esuite.n et.../...
```



- **Wireshark** is the world's foremost and widely-used network protocol analyzer
- It lets you see what's happening on your network

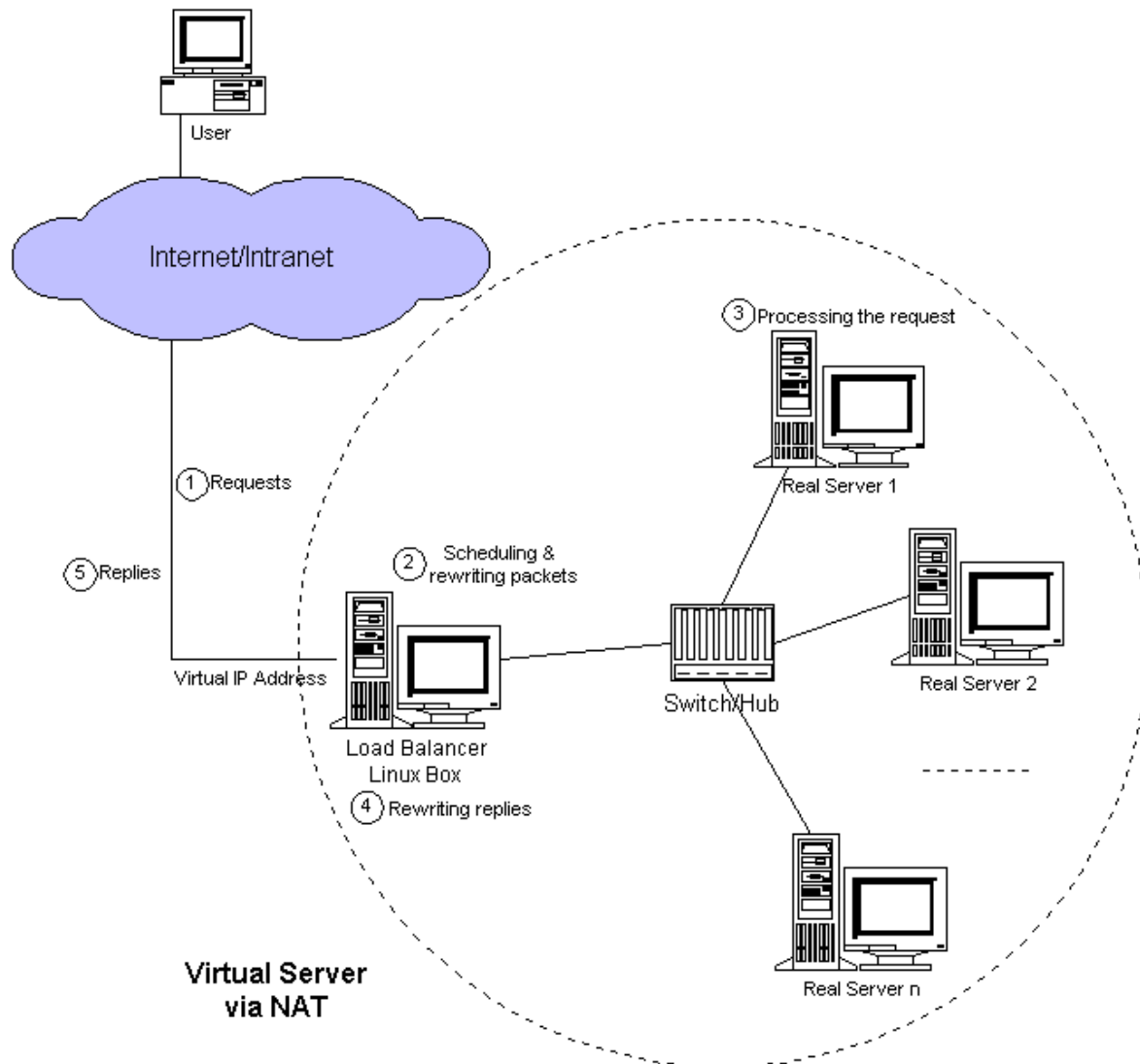
LVS modes



- **LVS can work at three different modes.**

<https://www.fatalerrors.org/a/lvs-keepalived-high-availability-cluster.html>

LVS – NAT mode



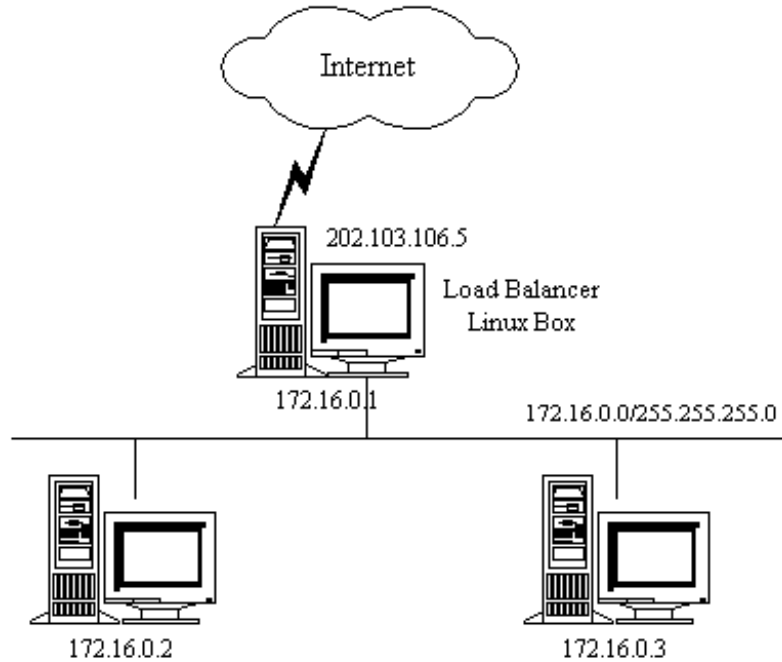
```
[root@lvs ~]# ipvsadm -e -t 172.16.9.100:80 -r 192.168.17.27 -m -w 3
[root@lvs ~]# ipvsadm -e -t 172.16.9.100:80 -r 192.168.17.37 -m -w 6
[root@lvs ~]# ipvsadm -Ln
IP Virtual Server version 1.2.1 (size=4096)
Prot LocalAddress:Port Scheduler Flags
  -> RemoteAddress:Port          Forward Weight ActiveConn InActConn
TCP  172.16.9.100:80 wrr
  -> 192.168.17.27:80           Masq    3      0      28
  -> 192.168.17.37:80           Masq    6      0      25
```

- “ipvsadm” is the tool for configuration LVS
- NAT means **Network Address Translation**
- RS (real server) need to set LVS server as gateway
- Both request and response pass LVS server

<https://zhuanlan.zhihu.com/p/71825940>

https://blog.csdn.net/weixin_44800915/article/details/107034112

LVS – NAT mode



Protocol	Virtual IP Address	Port	Real IP Address	Port	Weight
TCP	202.103.106.5	80	172.16.0.2	80	1
			172.16.0.3	8000	2
TCP	202.103.106.5	21	172.16.0.3	21	1

Packet rewriting works as follows.

The incoming packet for web service would have source and destination addresses as:

SOURCE 202.100.1.2:3456 DEST 202.103.106.5:80

The load balancer will choose a real server, e.g. 172.16.0.3:8000. The packet would be rewritten and forwarded to the server as:

SOURCE 202.100.1.2:3456 DEST 172.16.0.3:8000

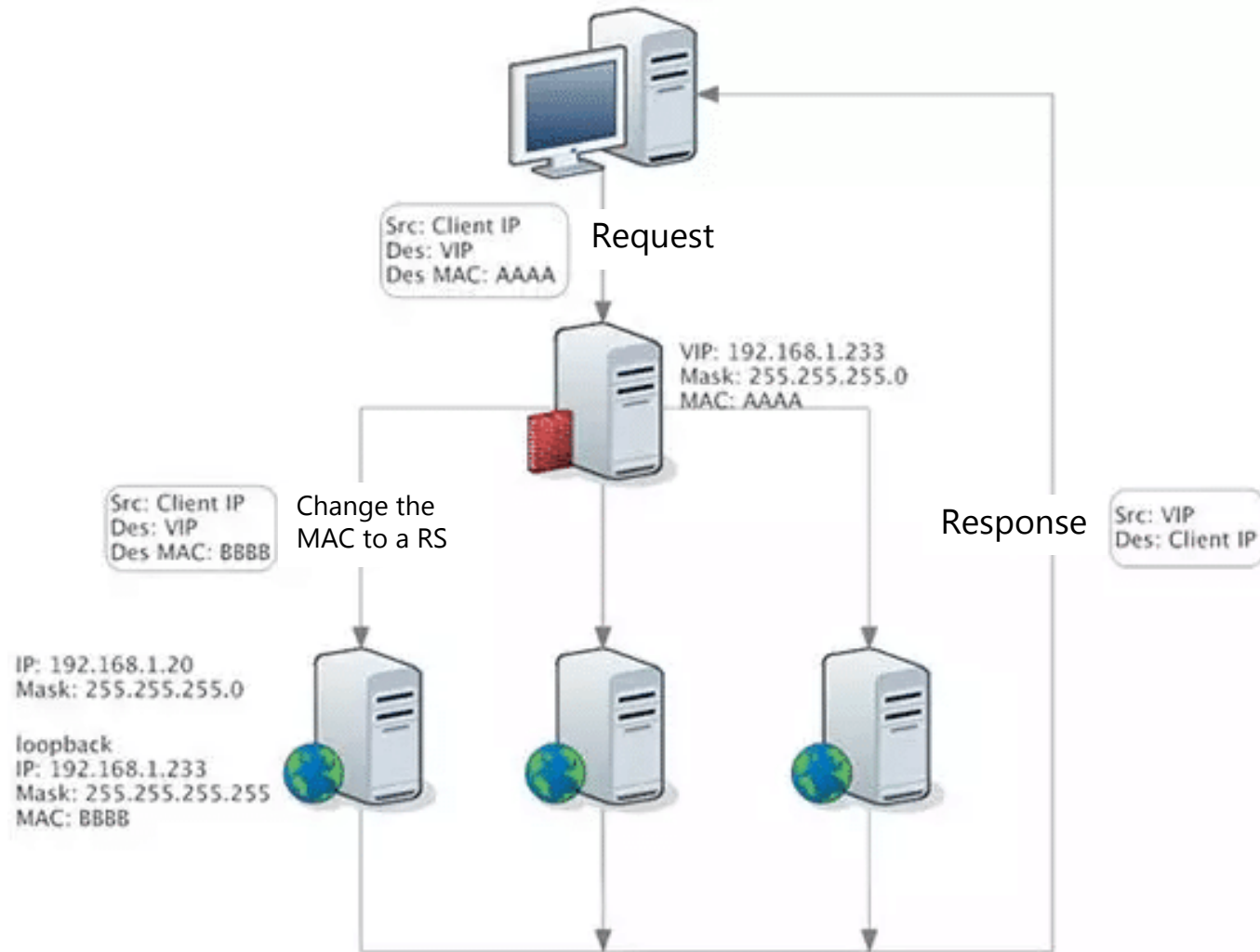
Replies get back to the load balancer as:

SOURCE 172.16.0.3:8000 DEST 202.100.1.2:3456

The packets would be written back to the virtual server address and returned to the client as:

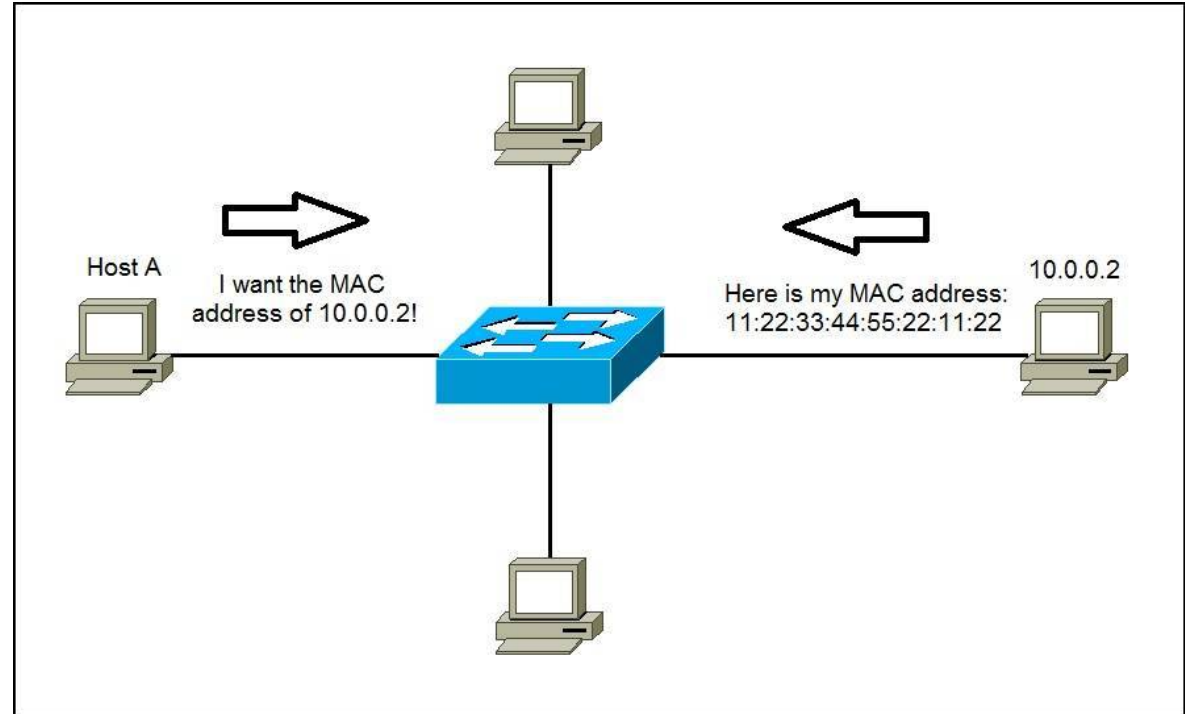
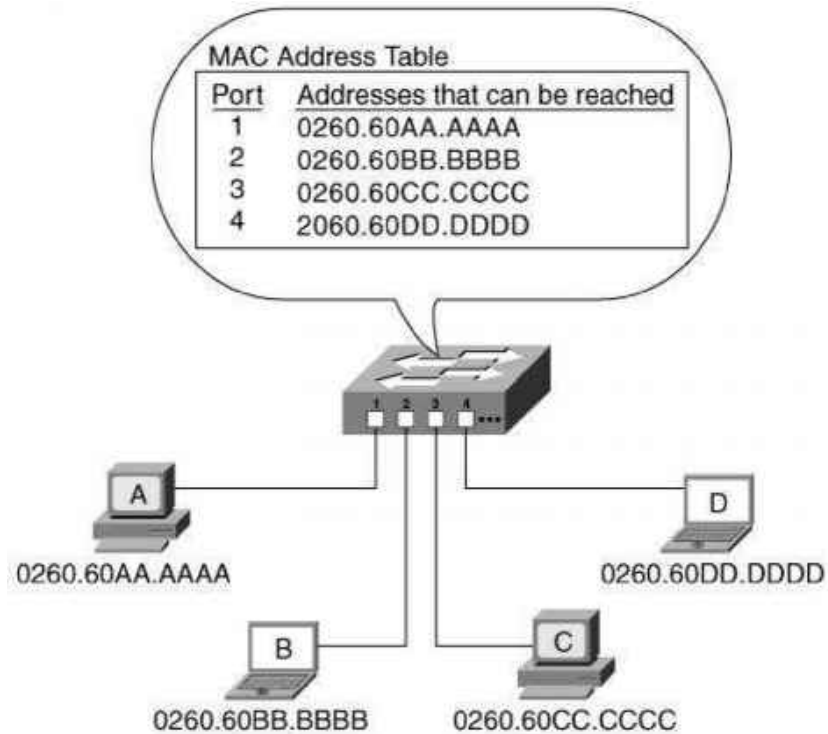
SOURCE 202.103.106.5:80 DEST 202.100.1.2:3456

LVS – DR mode



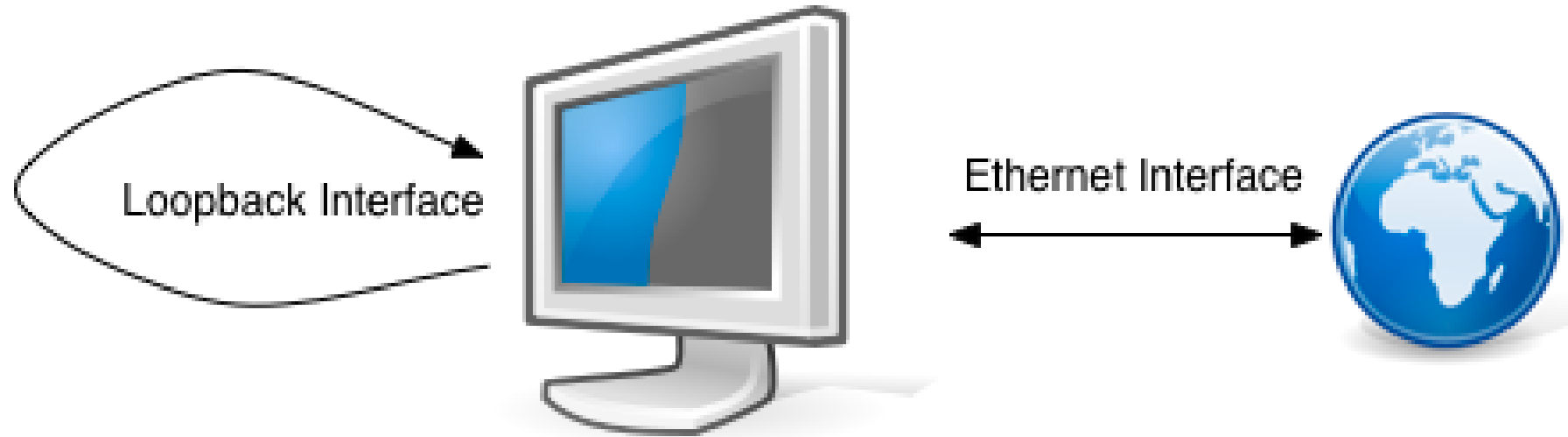
- DR means **Direct Route**
- LVS computer only “play” the request packages and RS directly reply end-user
- Normally, the request packages are small, the response packages are big
- This configuration can further improve the performance

LVS – Switch & ARP



- How does Switch build up the MAC address table?

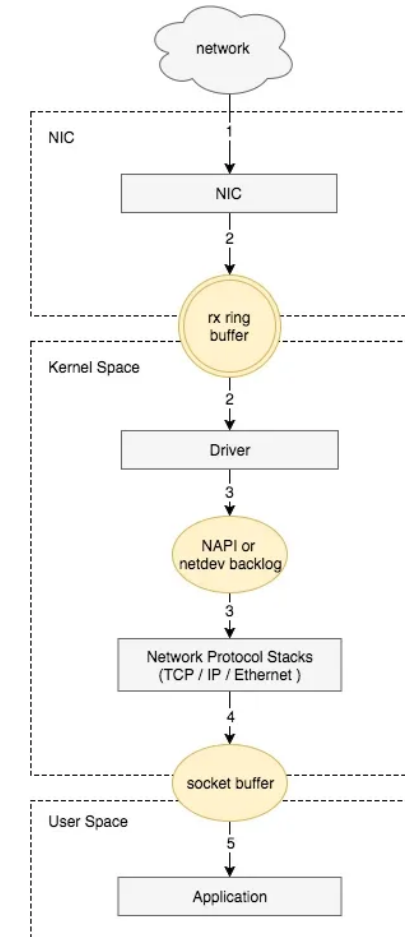
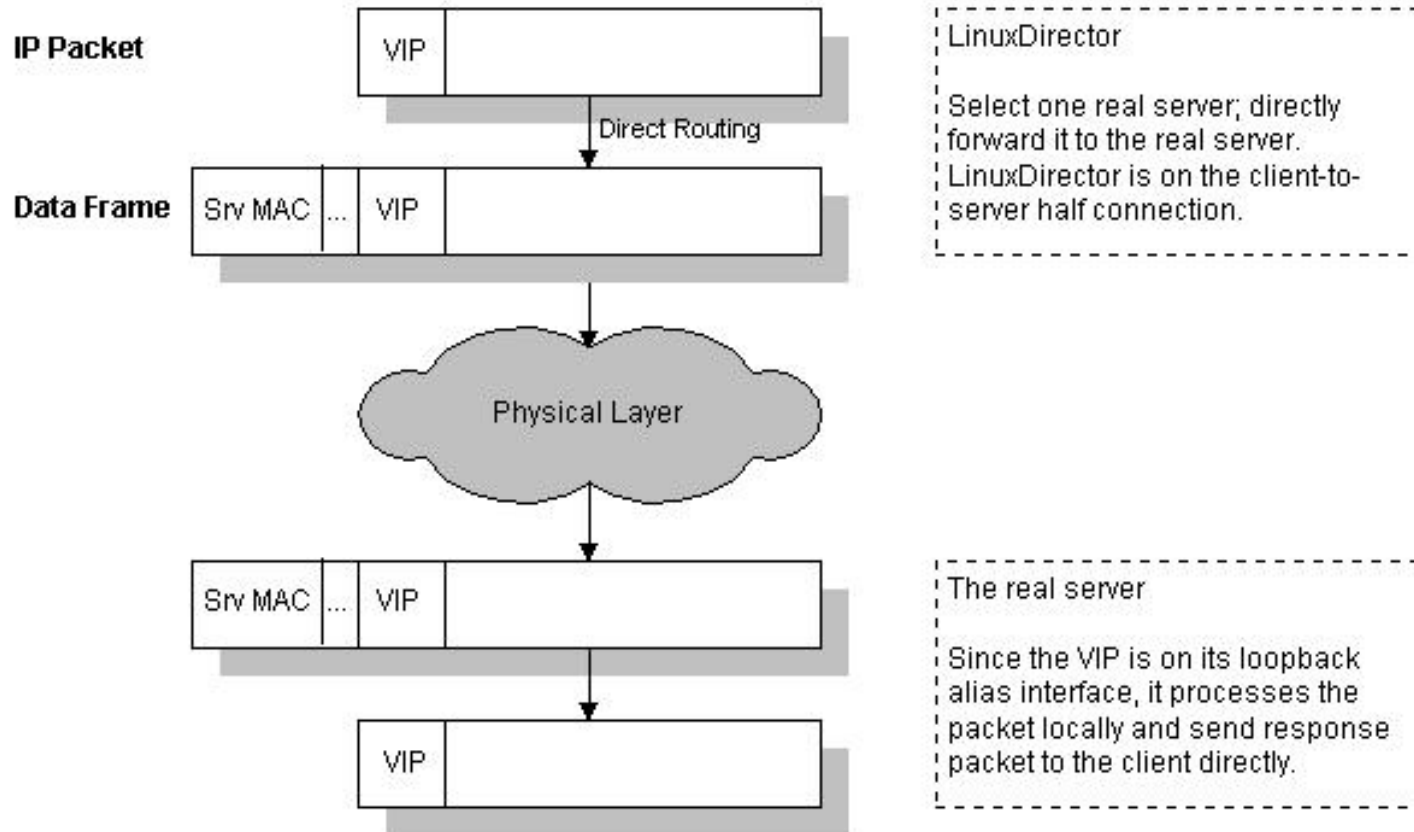
LVS - localhost



- The normal network card is called an Ethernet device
- Loopback is an "virtual" network card type device.
- **Every traffic that you send to loopback will come back.**

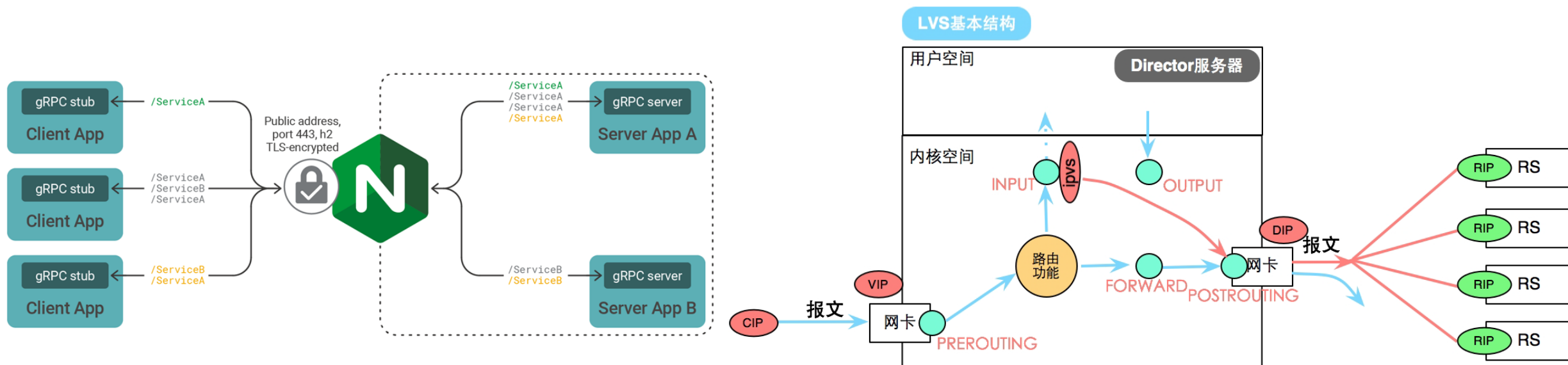
<https://superuser.com/questions/321734/how-does-localhost-127-0-0-1-work>

LVS – DR is a kind of “cheating”



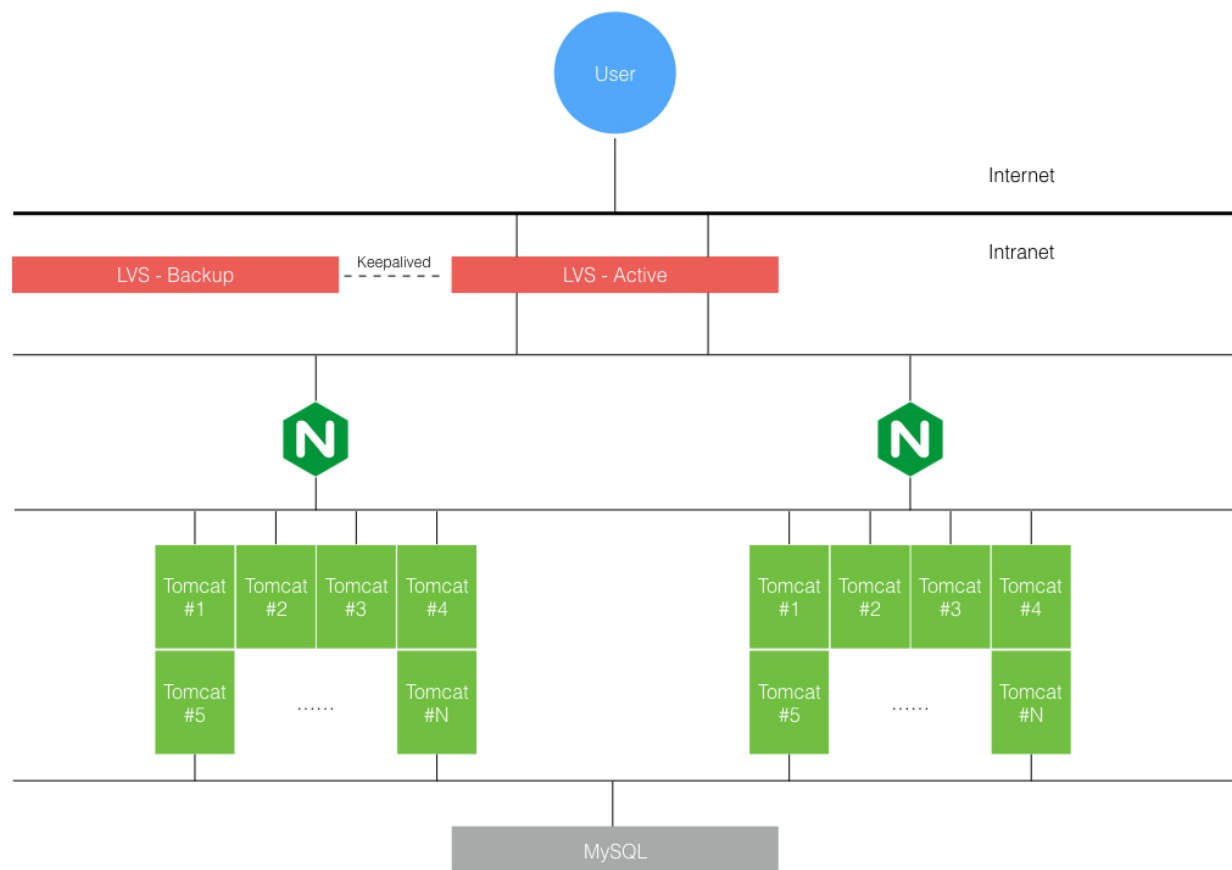
- Real server need to turn off ARP functions

Nginx vs. LVS



- **Choose Which one?**
- They focus on different level: HTTP vs. TCP
- Nginx can handle the URL/HTTP request information
- LVS is much fast since its only process the TCP part

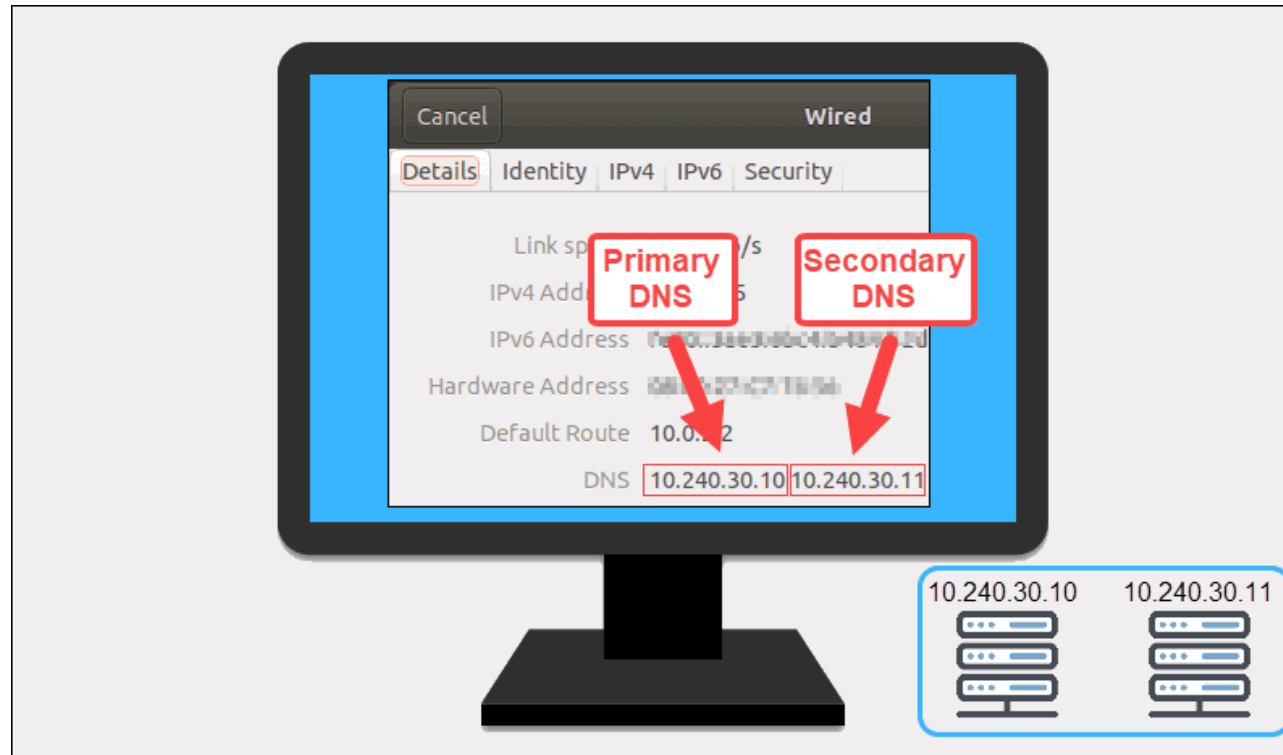
Best practice: LVS + Nginx



- Normally, the LVS and Nginx are used together at different levels
- Load balance also need "keepalive"
- Reported benchmark for a cluster:
 - 10Gb/s flow
 - 1,000,000 QPS

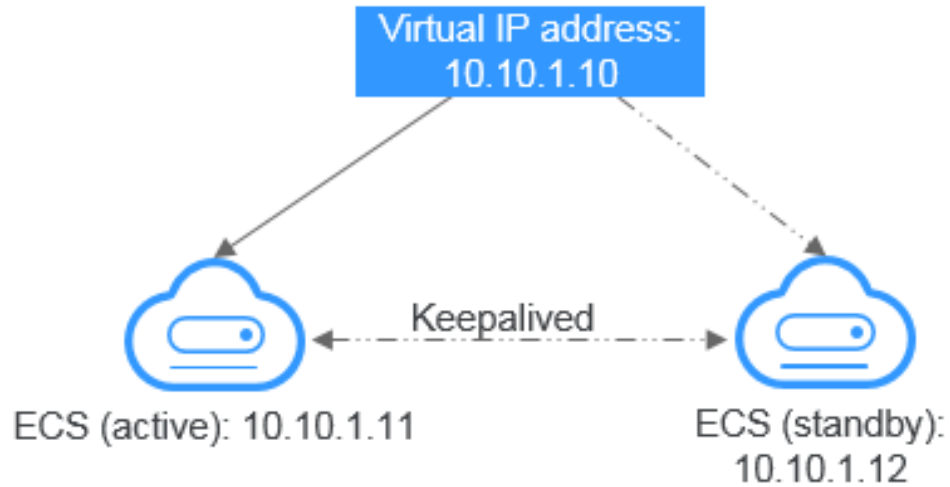
<https://www.zhihu.com/question/21237968>

Host standby



- Two "servers" has the same function.
- If the primary one is gone, switch to the secondary one by the client.

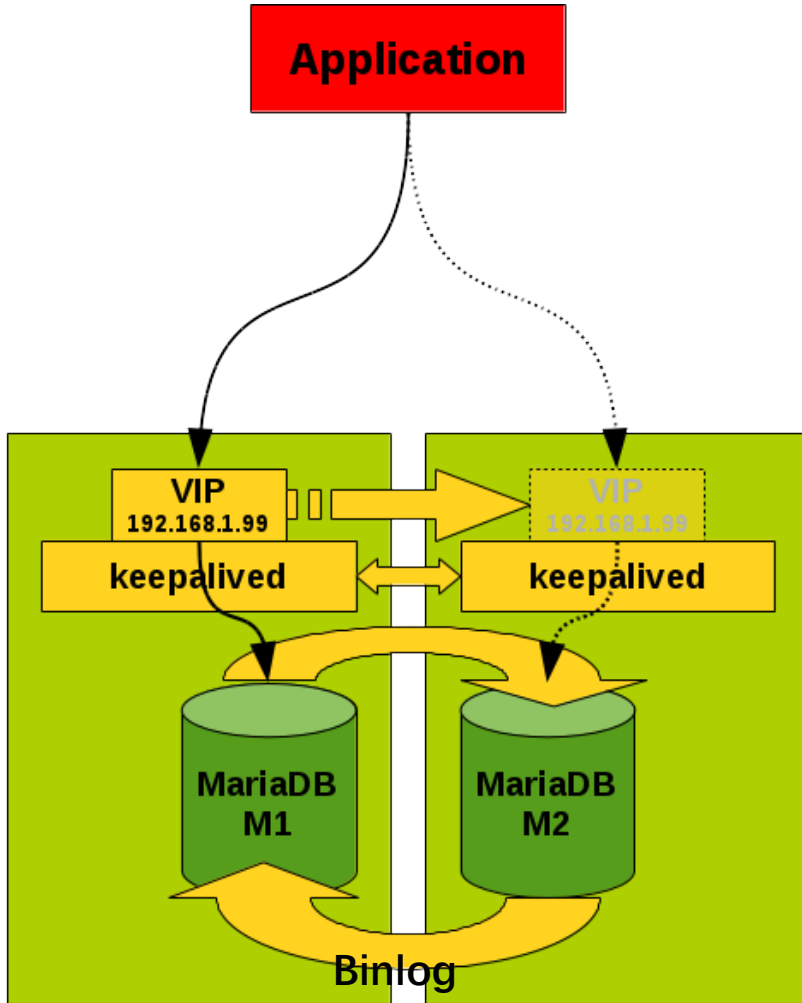
VRRP (Virtual Routing Redundancy Protocol)



- Two “servers” has the same configuration.
- Standby one would check the health of active one.
- Once the active one is gone, the standby one would take over the network traffic.
- A typical software for this function: **keepalived**

https://sc1.checkpoint.com/documents/R76/CP_R76_Gaia_WebAdmin/87911.htm

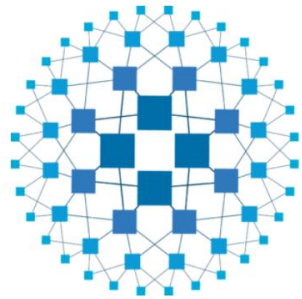
Dual master MySQL



```
[root@vm-002 mysql]# mysqlbinlog mysql-bin.000002
.....
# at 624
#160925 21:29:53 server id 1 end_log_pos 796 Query thread_id=3 exec_time=0 error_code=0
SET TIMESTAMP=1474810193/*!*/;
insert into member(` name`,` sex`,` age`,` classid`) values('wangshibo','m',27,'cls1'),
('guohuihui','w',27,'cls2')      #执行的sql语句
/*!*/;
# at 796
#160925 21:29:53 server id 1 end_log_pos 823 Xid = 17      #执行的时间
.....
```

- Binlog is the original commands applied to DB.
- Binlog was used to replicate the data.
- No contribution to the con-currency performance.
- During the switching, it would face the data **inconsistency** problem.

Other choices: HA proxy and f5



HAPROXY

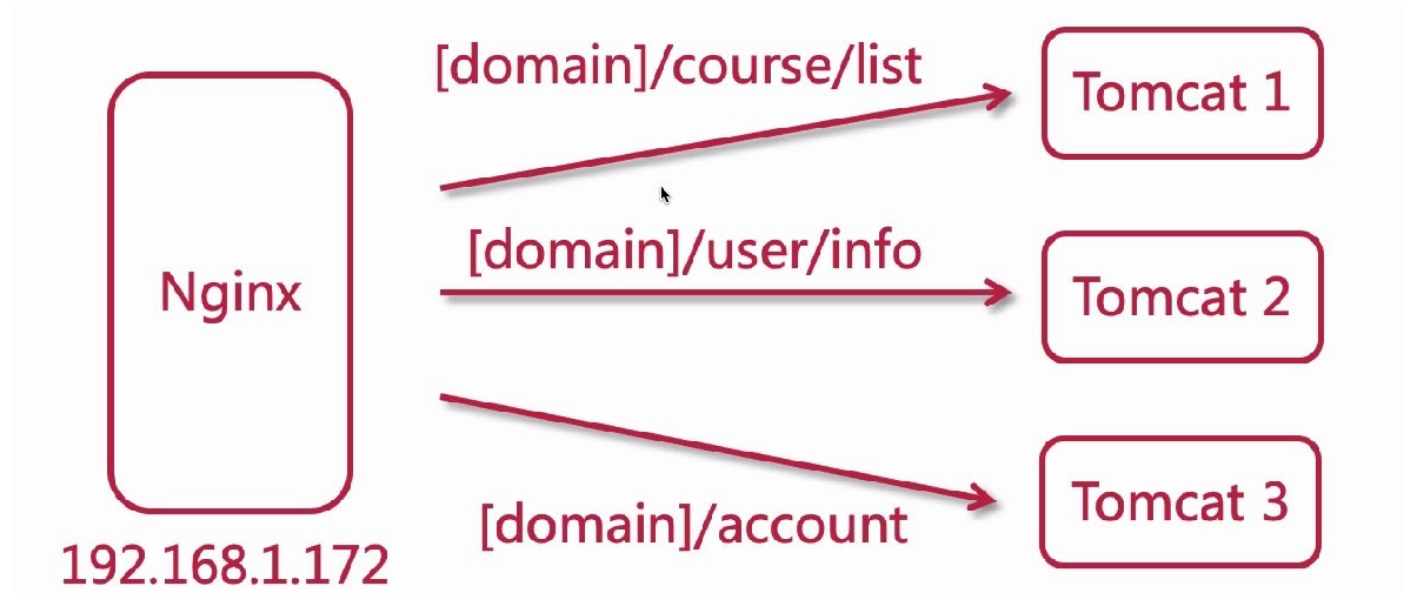
- **Another popular proxy software**
- **Support both layer 4 and layer 7**



- **Optimized hardware for load balance**
- **Higher performance on the cost of high price**

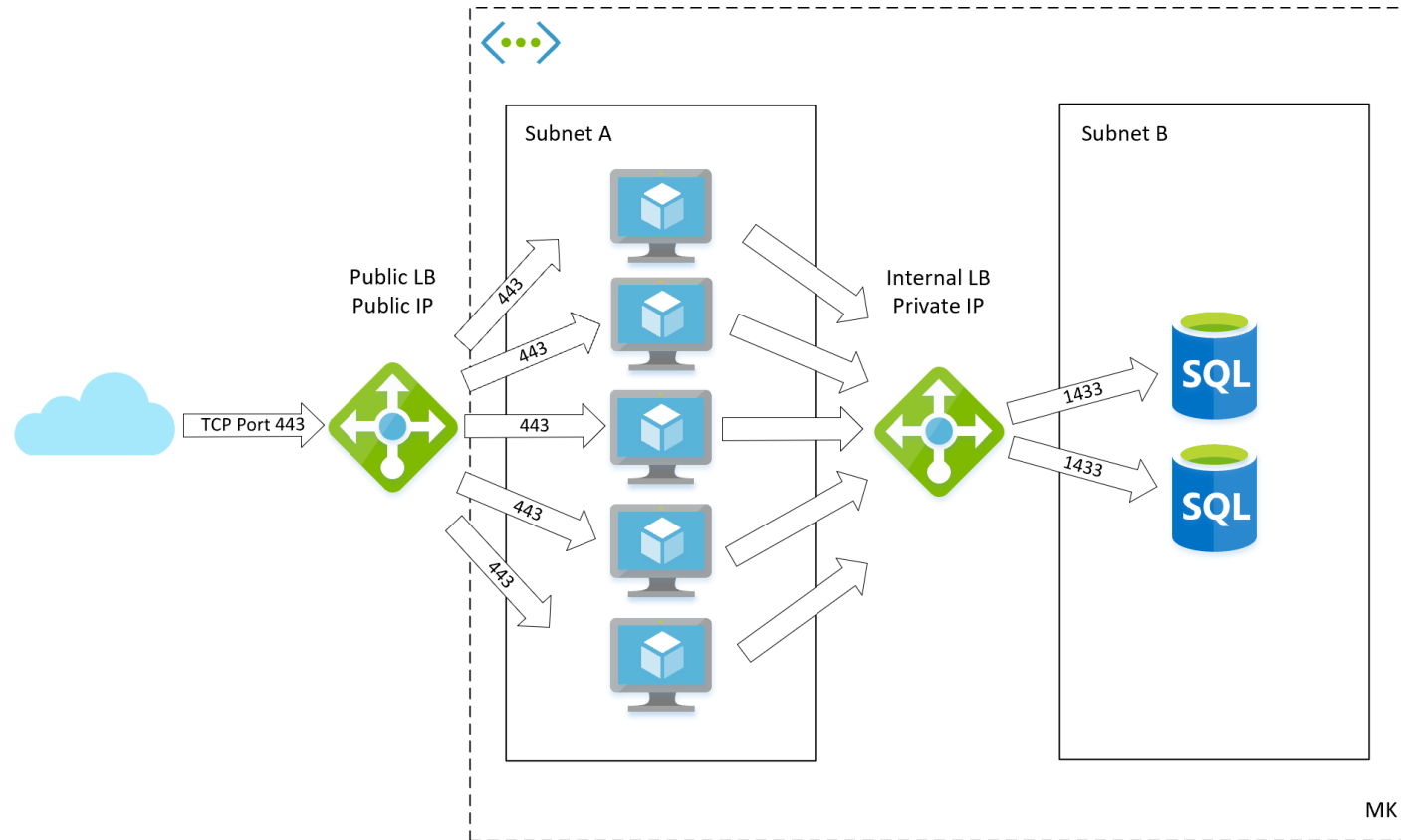
Balancing techniques

```
server {  
    listen      80 default_server;  
    listen     [::]:80 default_server;  
    server_name _;  
    root /data/web/h5/dist;  
    # root      /usr/share/nginx/html;  
    # Load configuration files for the default server block.  
    include /etc/nginx/default.d/*.conf;  
  
    1 location / {  
        root /data/web/h5/dist;  
    }  
    location ^~/img/ {  
        root /data;  
    }  
  
    2 location ^~/api/{  
        # proxy_pass http://5hv33u.natappfree.cc;  
        proxy_pass http://localhost:7500;  
    }  
  
    location ^~/app/{  
        # proxy_pass http://5hv33u.natappfree.cc;  
        proxy_pass http://localhost:6668;  
    }  
  
    3 location /backend/{  
        # proxy_pass http://5hv33u.natappfree.cc;  
        proxy_pass http://localhost:8016/;  
    }  
  
    location ^~/zyj/{  
        # proxy_pass http://5hv33u.natappfree.cc;  
        proxy_pass http://localhost:6665;  
    }  
}
```



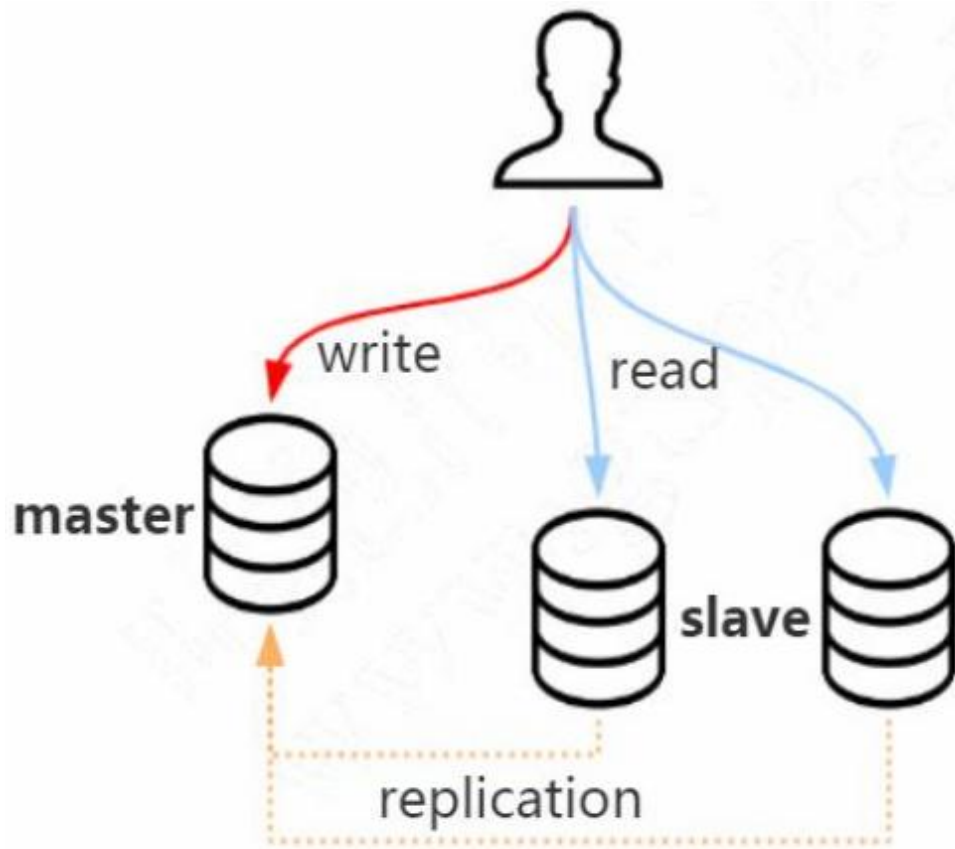
- Business level load balancing: Route requests to different vertical clusters based on URL paths (such as /user*/, /order/).

The new bottleneck



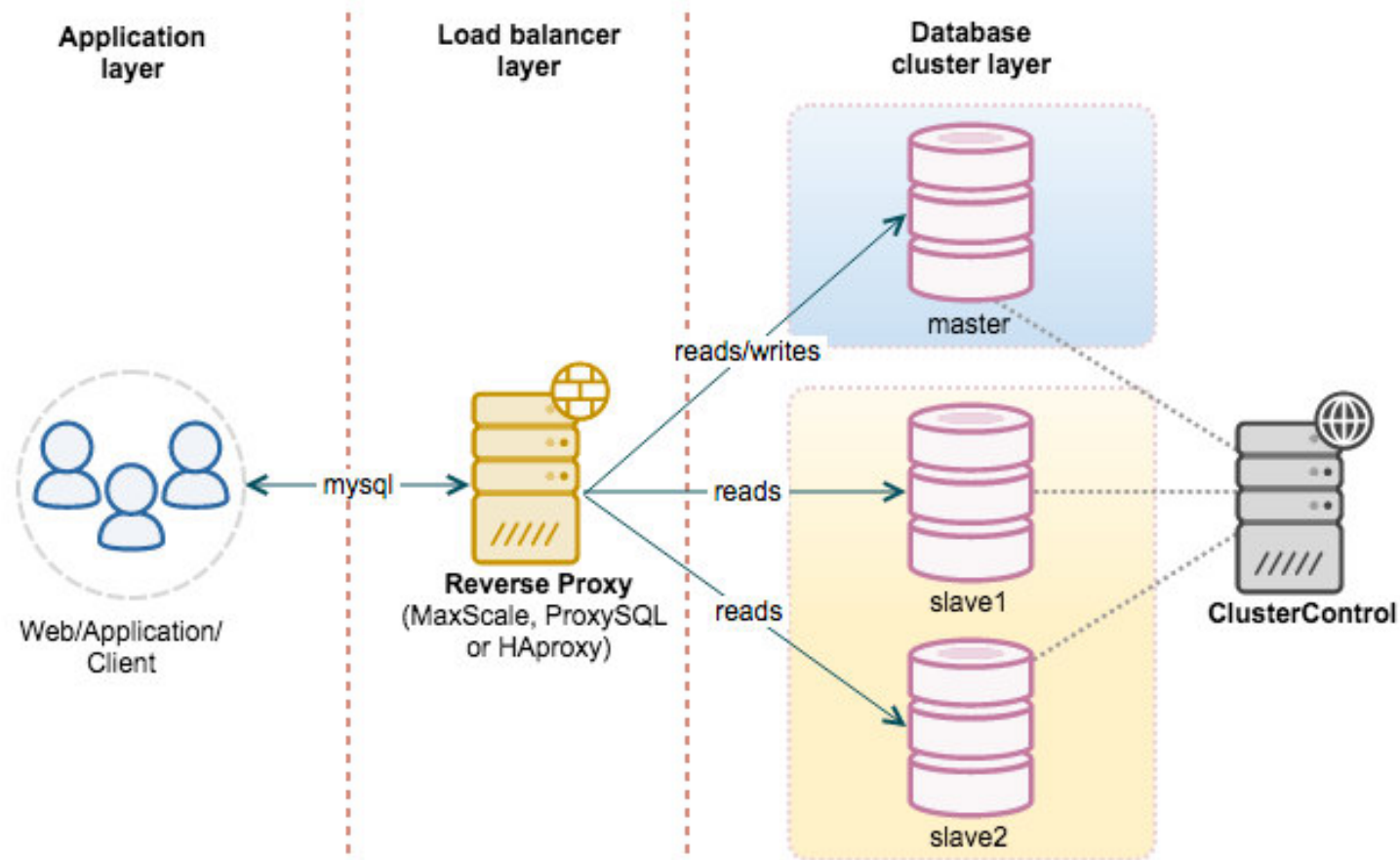
- The database may become the bottleneck for the whole system.
- Database also needs **load balance**.

Master/Slave MySQL



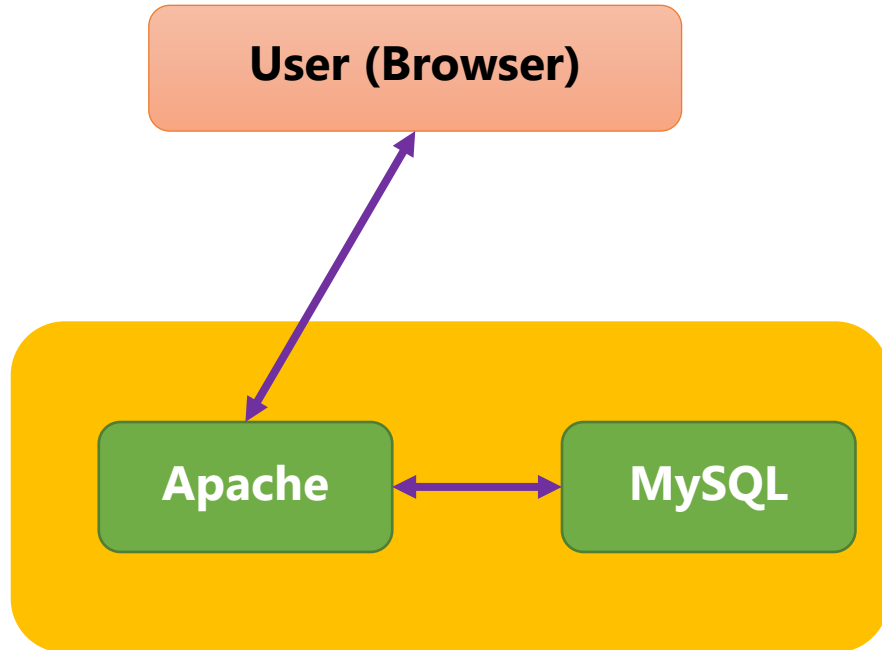
- For e-commerce application, "read" (i.e. search) takes a very large part of all requests.
- The slave DB only support "read".
- Only write would apply to master DB and more than one server to support the read. The "con-currency" capability would be extended.
- If the HA is still needed, the master should be "dual master".
- The application need to distinguish the write and read in their code. It is not very friendly.

Proxy the DB cluster



- Let the proxy smart choose the master or slave.
- If the proxy server is broken... We need to have a **"keepalive"** proxy server.

Our mini-web

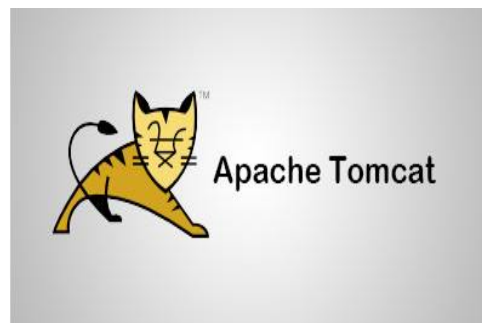
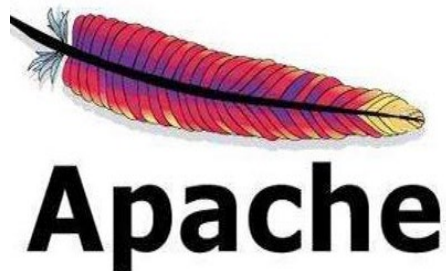


23094832	23355647	94643456	48355551	23094832	46478126	55280457	50845634
98564875	65861111	48916498	11001018	98564875	17823156	56788832	32535466
56732149	87484647	68456343	64578956	56732149	73213321	13321672	89355644
0-932476	81261782	25354668	87776886	0-932476	67223154	23154312	57577658
98345656	31115673	93556476	65478516	98345656	31221453	21453201	61111874
00874768	21332167	58647869	55844551	00874768	20113156	18831565	84647812
55647016	22315431	68866874	61494643	55647016	54345874	43577458	61743122
89355647	22145321	84610478	45648964	89355647	98645789	74986457	14532131
65864786	31500065	12617823	98684563	65864786	56877768	74457458	50006565
96880006	6541					8777	43458749
68748464	4986					0654	86421315
78126178	1565					6744	65434587
23156732	8749					6149	49886845
13321672	7899					8564	63432535
23154312	76886654	77688665	12617823	23154312	84563432	89649860	46689355
21453201	78516558	47851655	15673213	21453201	53546689	84563432	64765864
13156543	44551614	84455161	32110672	13156543	35564765	53546689	78696886
45874986	94643456	49464345	64564868	45874986	86111187	35564457	68748461
45789568	48916498	64896498	57875867	45789568	48464781	57765861	04781261
77768860	68456343	68456343	88944334	77768860	26178231	11187484	78231567
65478516	25354668	25354668	21010011	65478516	11567321	64781261	32133216
55844551	93556476	93556476	56456475	55844551	33216722	78231115	72213156
06149464	58647869	58647869	61249765	06149464	31543122	67321332	54345874
34564896	68866874	68866874	91093485	34564896	14532131	16727231	98645789
49860845	84610478	84647812	39201010	49860845	50006565	54312214	56877768
63432535	12617823	61782315	10048924	63432535	43458749	53213150	86654785

OVERLOAD

- In our labs, web server and dB server are running on the same computer.
- Both Web server and DB software (i.e. MySQL) cost hardware resources.
- Once many users visiting the web concurrently

Typical single web servers' capability



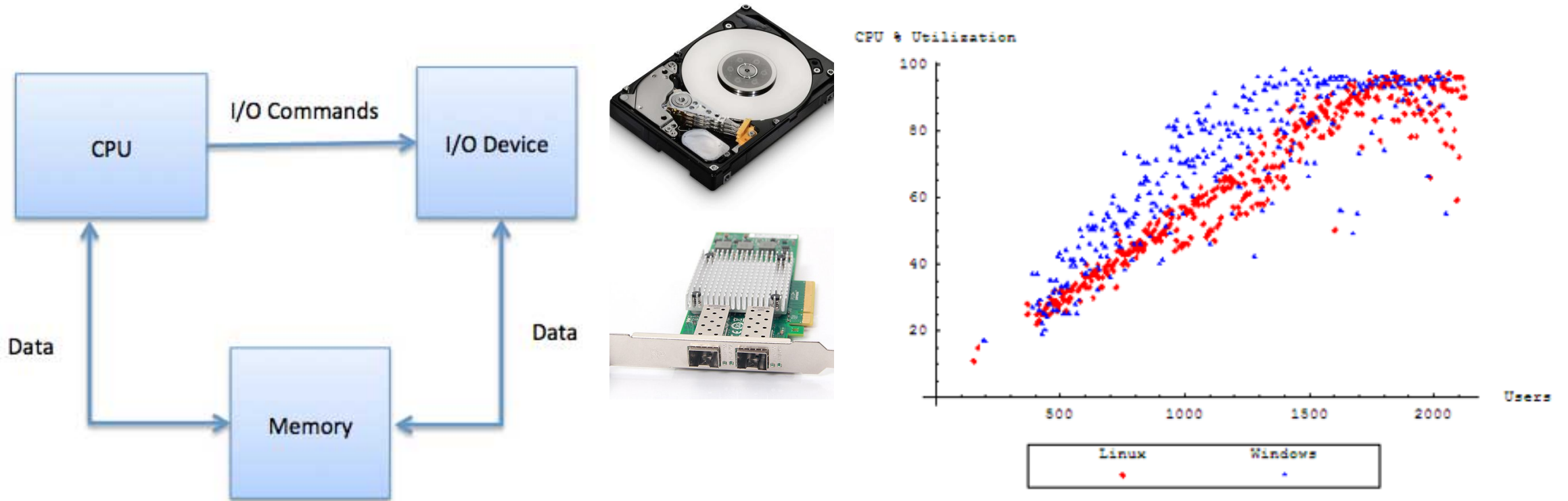
Typical concurrency:

- Tomcat 300
- Apache 3000
- Nginx 30000



<https://www.zhihu.com/question/19571087>
<https://www.jianshu.com/p/5198359b59f7>

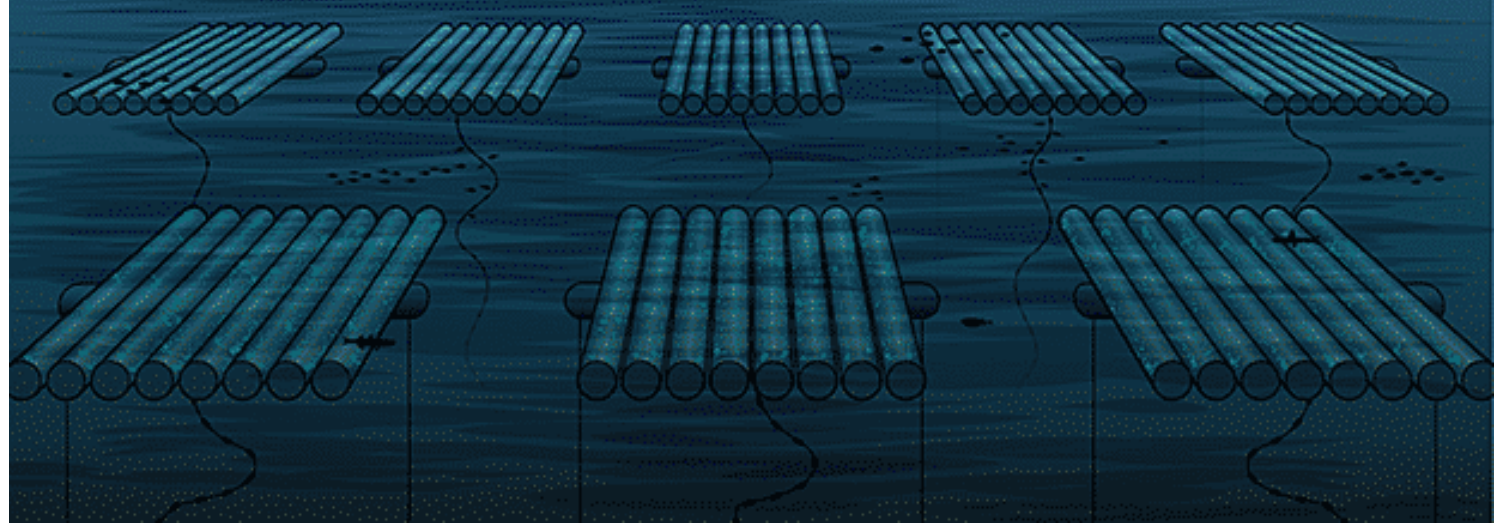
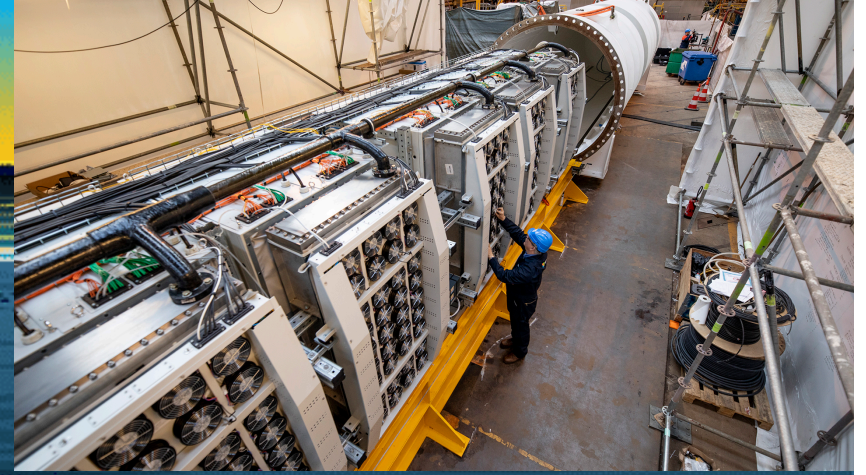
Each request costs resources



- Simply, no single computer can handle the 12306's traffic

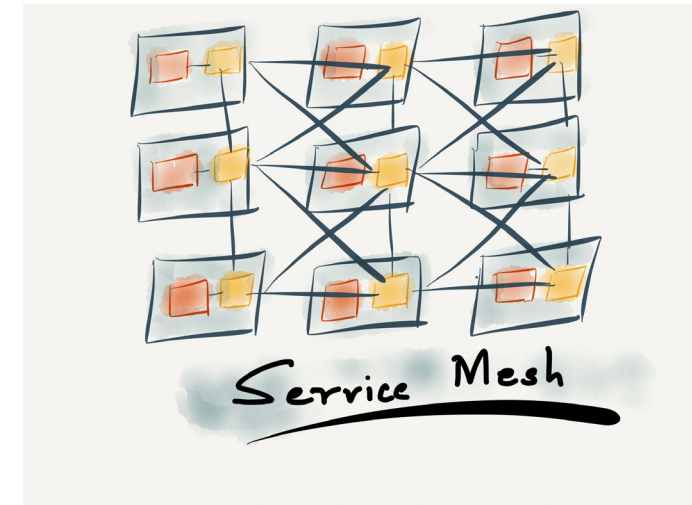
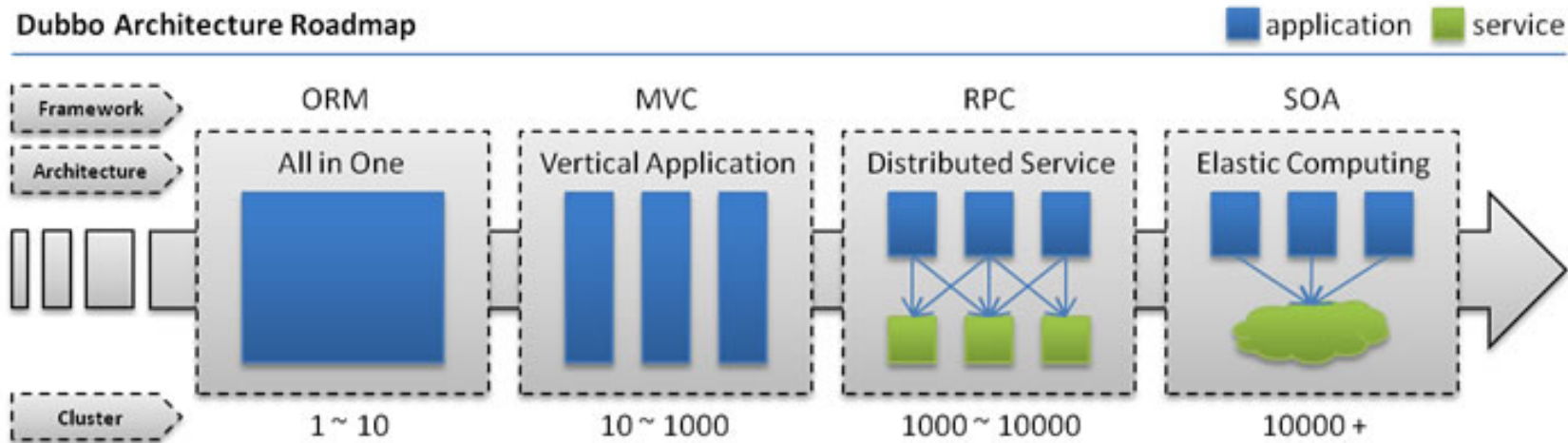
https://www.webperformance.com/library/reports/windows_vs_linux_part2/

Hero or a troop?



- A hero like a super power single computer.
- But we need to let lots of soldiers work together!

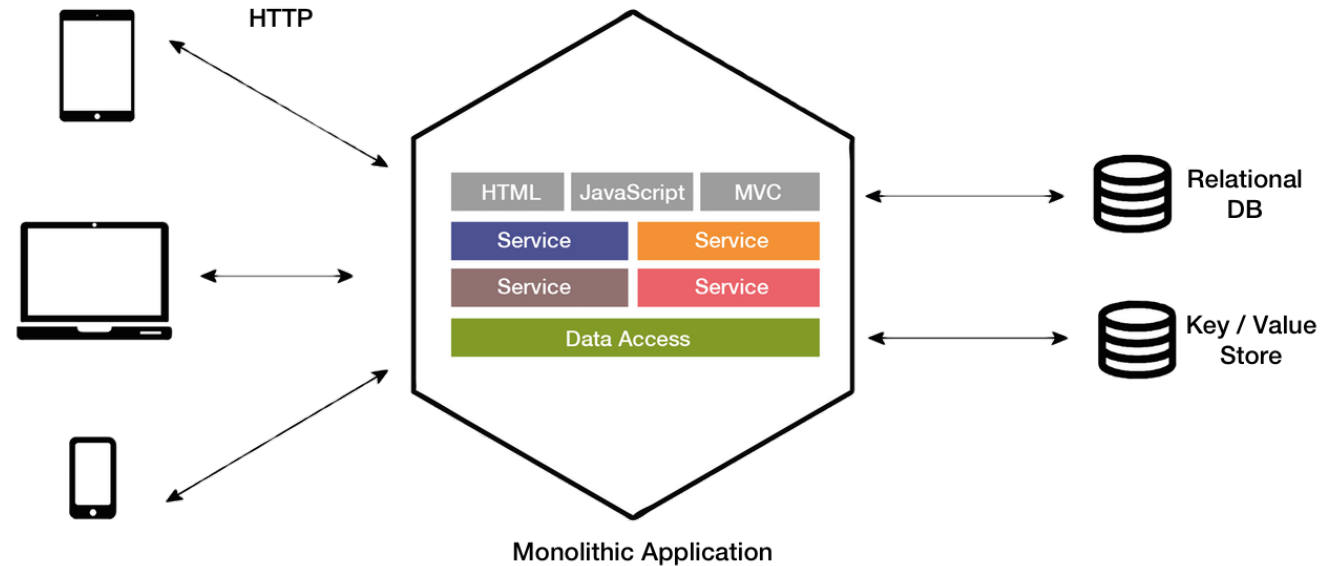
From one to pieces



- To operate the huge clusters, the project need to be split to tiny pieces

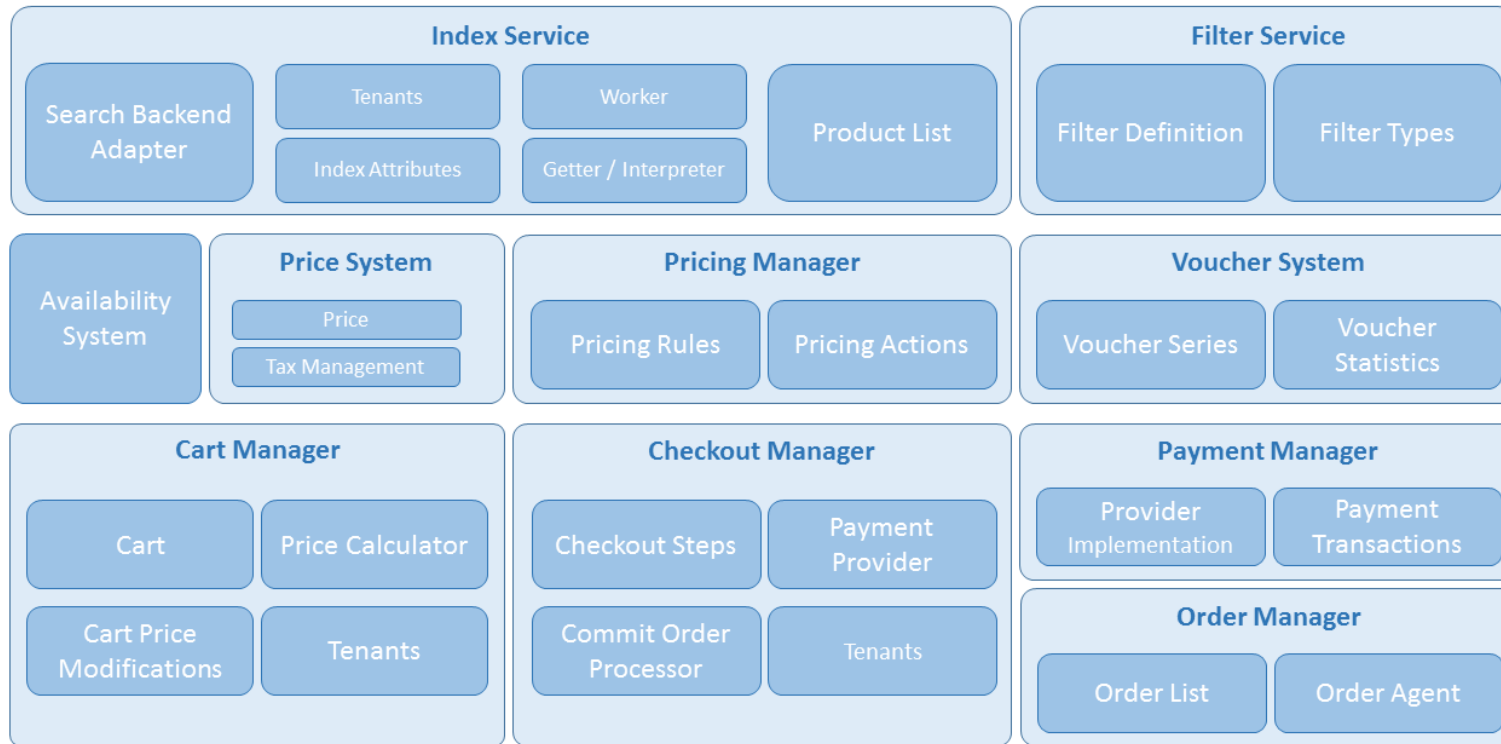
<https://dubbo.apache.org/en/docs/v2.7/user/preface/background/>

Monolithic architecture



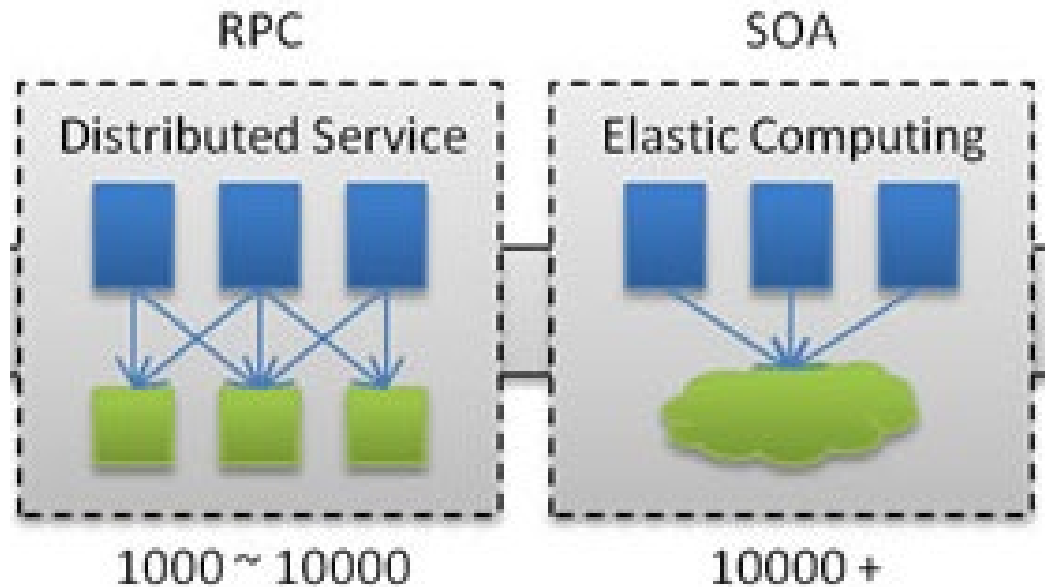
- When the traffic is very low, there is only one application, all the features are deployed together to reduce the deployment node and cost.
- At this point, the data access framework (ORM) is the key to simplifying the workload of the CRUD.
- **Load balancing** can be used for scaling application instances to cope with the growth of user request volume.

Vertical separation



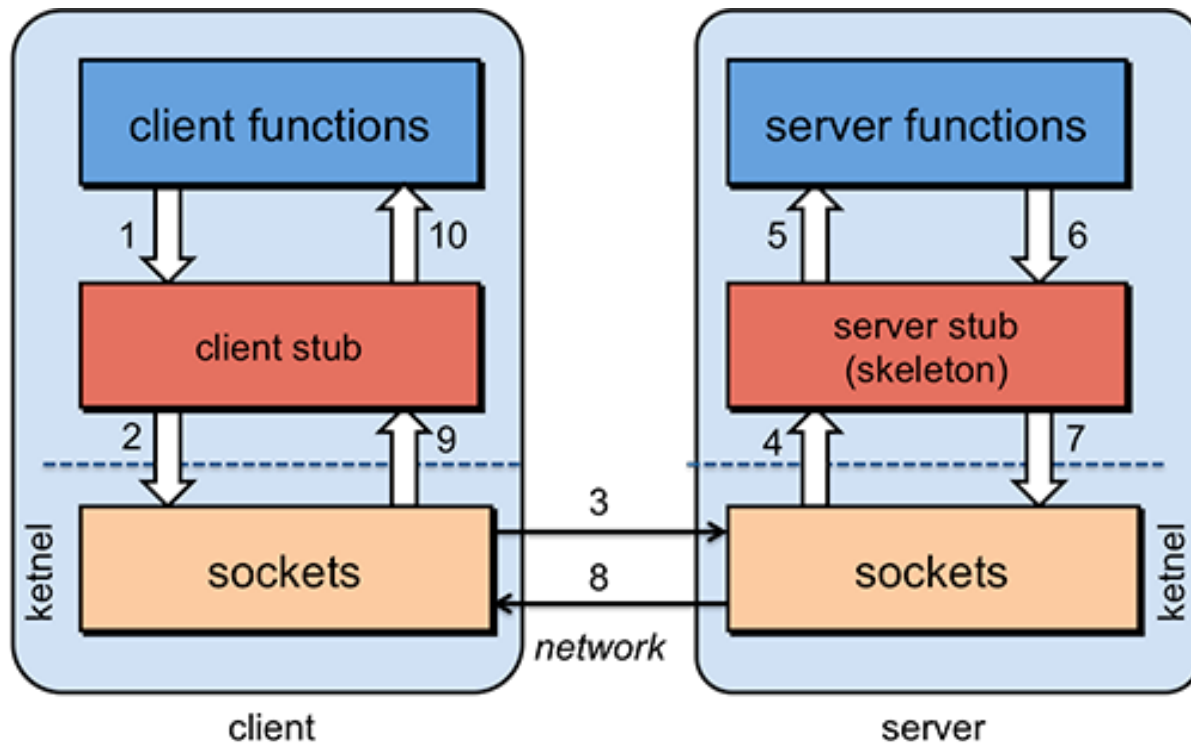
- When the traffic gets heavier, add monolithic application instances can not accelerate the access very well, one way to improve efficiency is to split the monolithic into discrete applications.
- Different functions can be encapsulated in the different components. Each components can be deployed as an individual cluster.

RPC and SOA



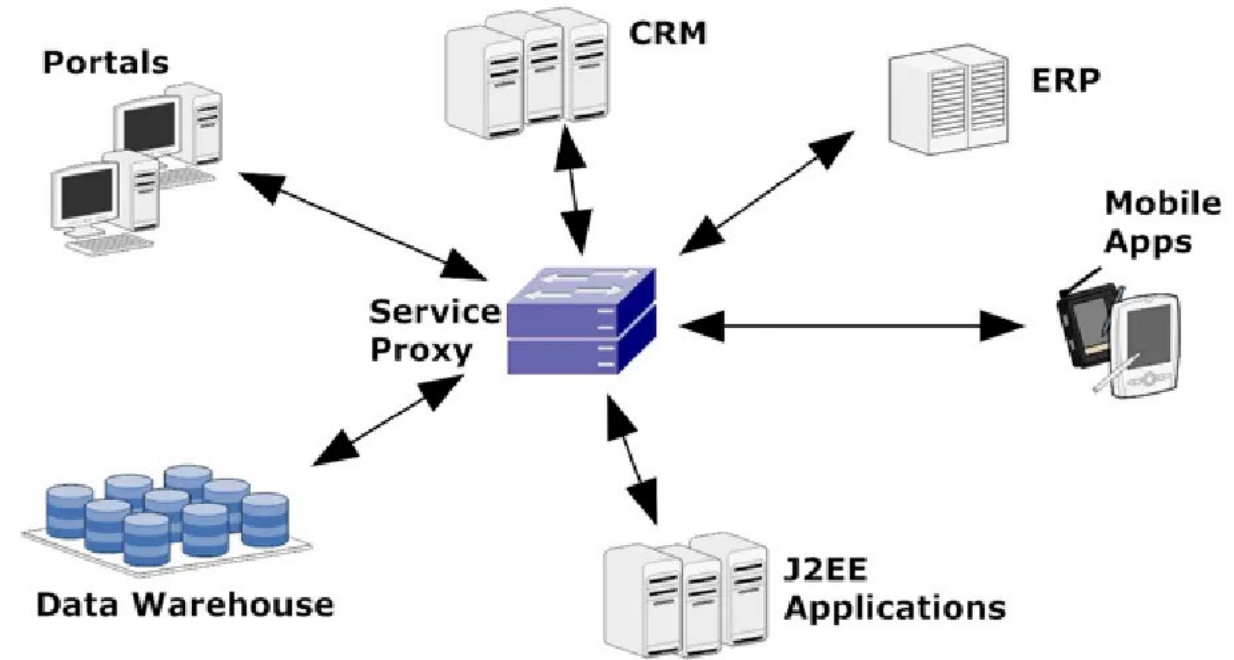
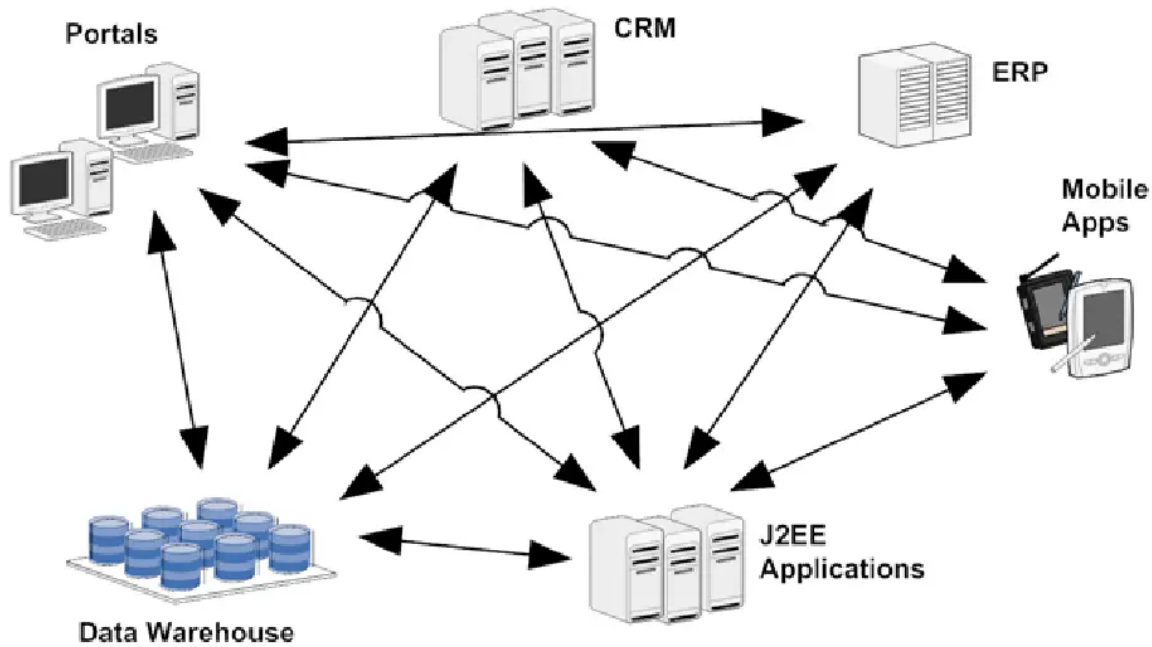
- When there are more and more vertical applications, the interaction between applications is inevitable, some core businesses (i.e. user) are extracted and served as independent services, which gradually forms a stable service center.
- This way the front-end application can respond to the changeable market demand more quickly.
- RPC and SOA are two different approaches.

RPC (Remote Procedure Call)



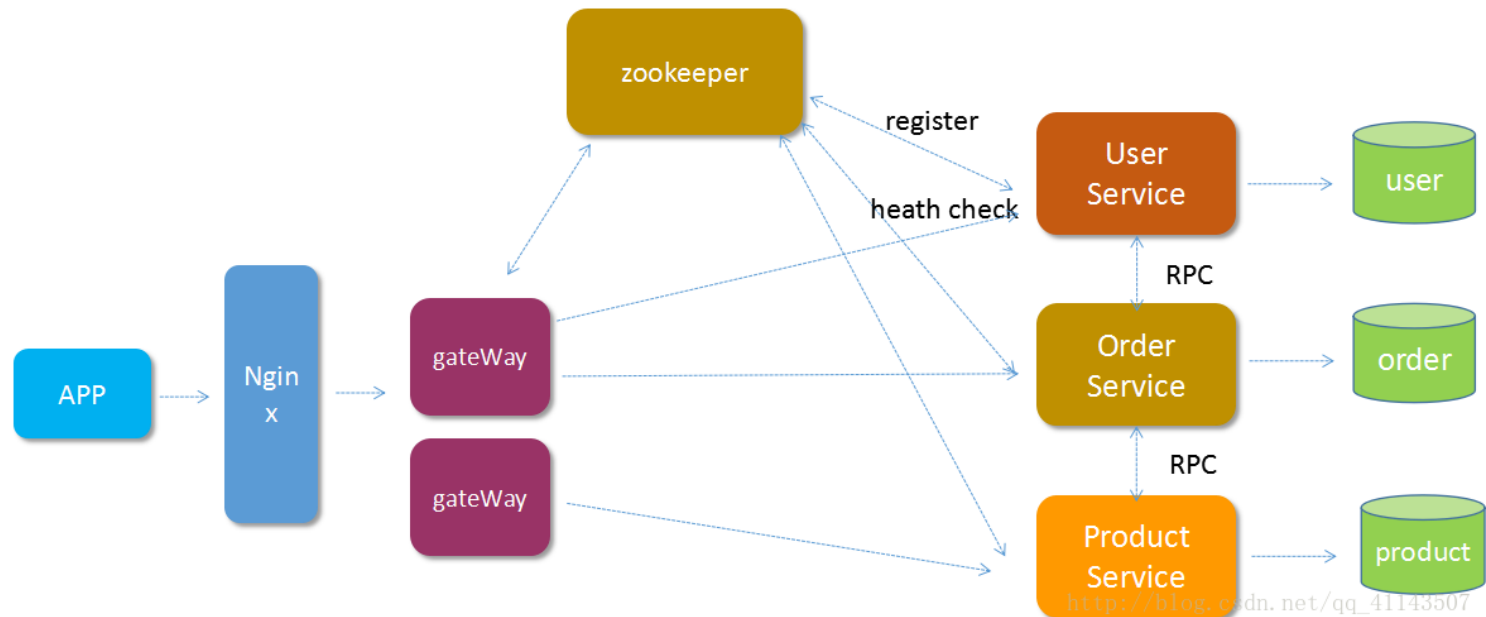
- In 1984, Birrell and Nelson devised a mechanism to allow programs to call procedures on other machines.
- Obviously, a remote procedure call is different from a local one in the underlying implementation.
- RPC can over TCP or HTTP layer.
- HTTP API is one kind of RPC.

A “router” is needed

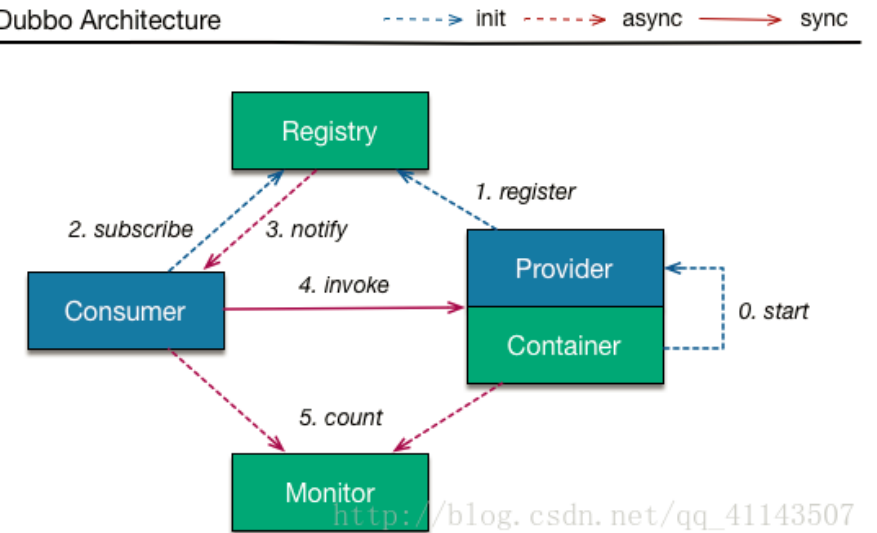


- When the point-point RPC calls become very complex
- An proxy can make them in order
- Let all communication through a router

Apache Dubbo



Dubbo Architecture

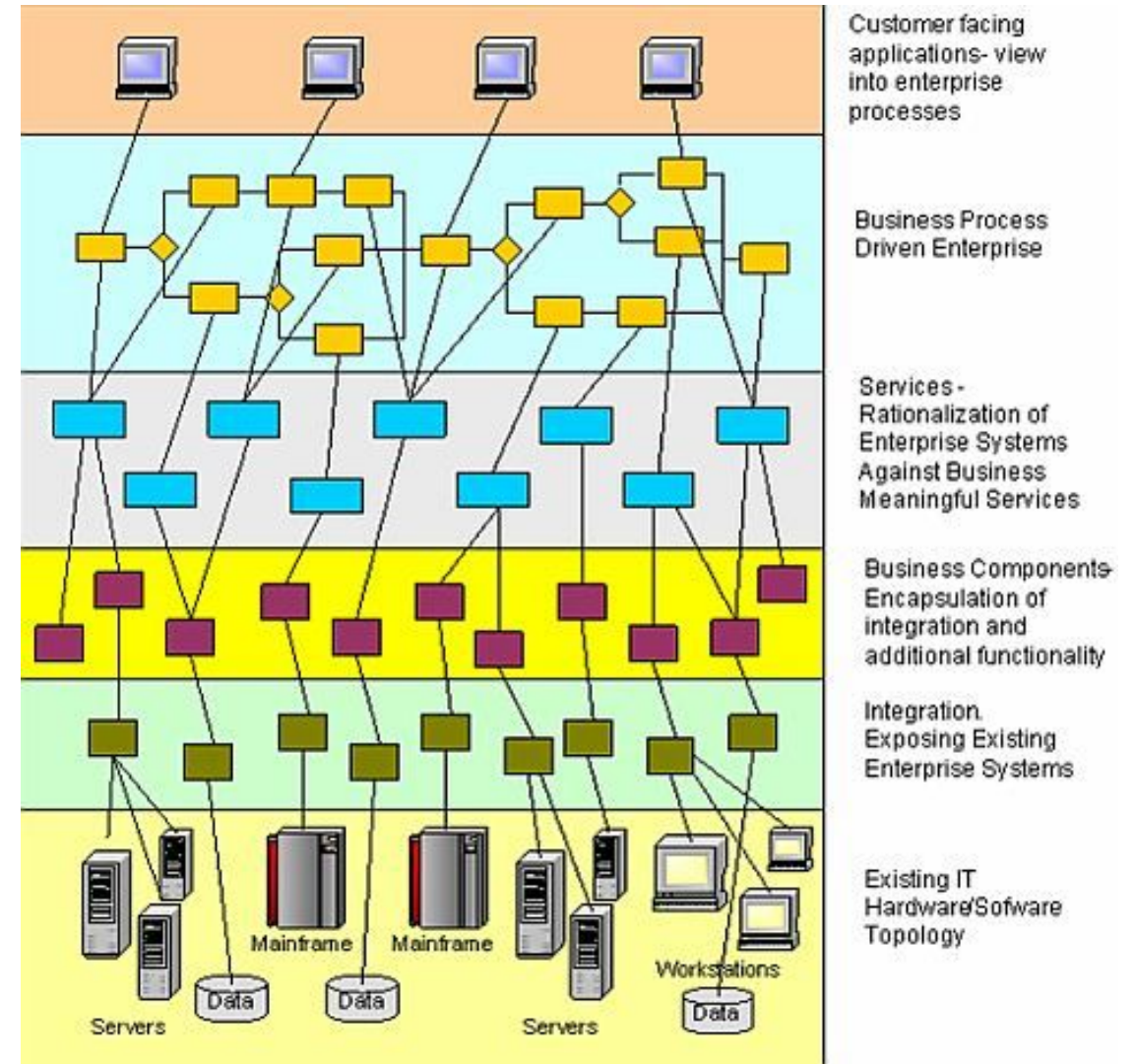


- Dubbo use zookeeper for registry and discovery
- It uses a gateway to provide the service to outside world.

https://www.sohu.com/a/206889245_115128

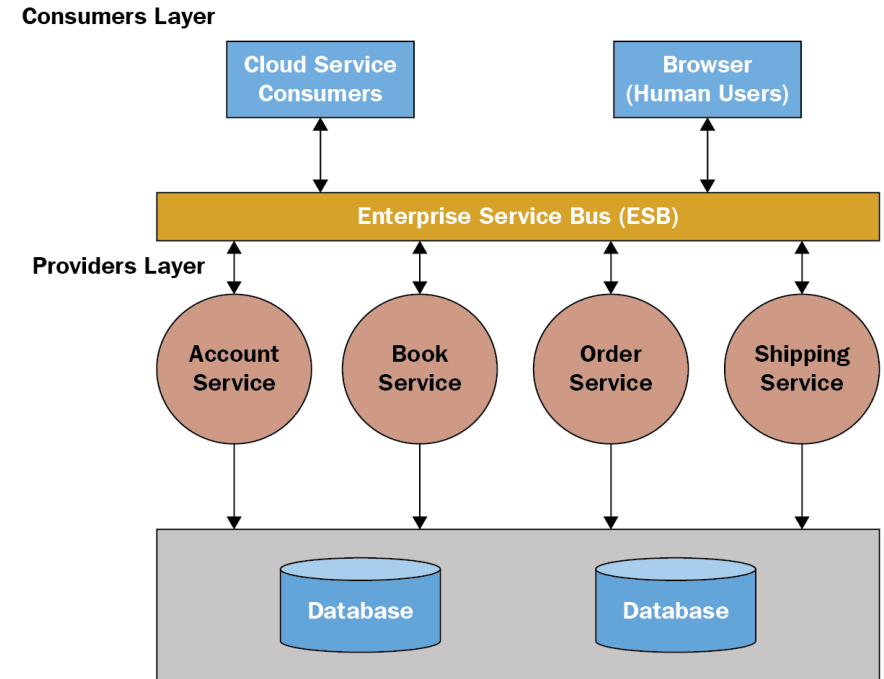
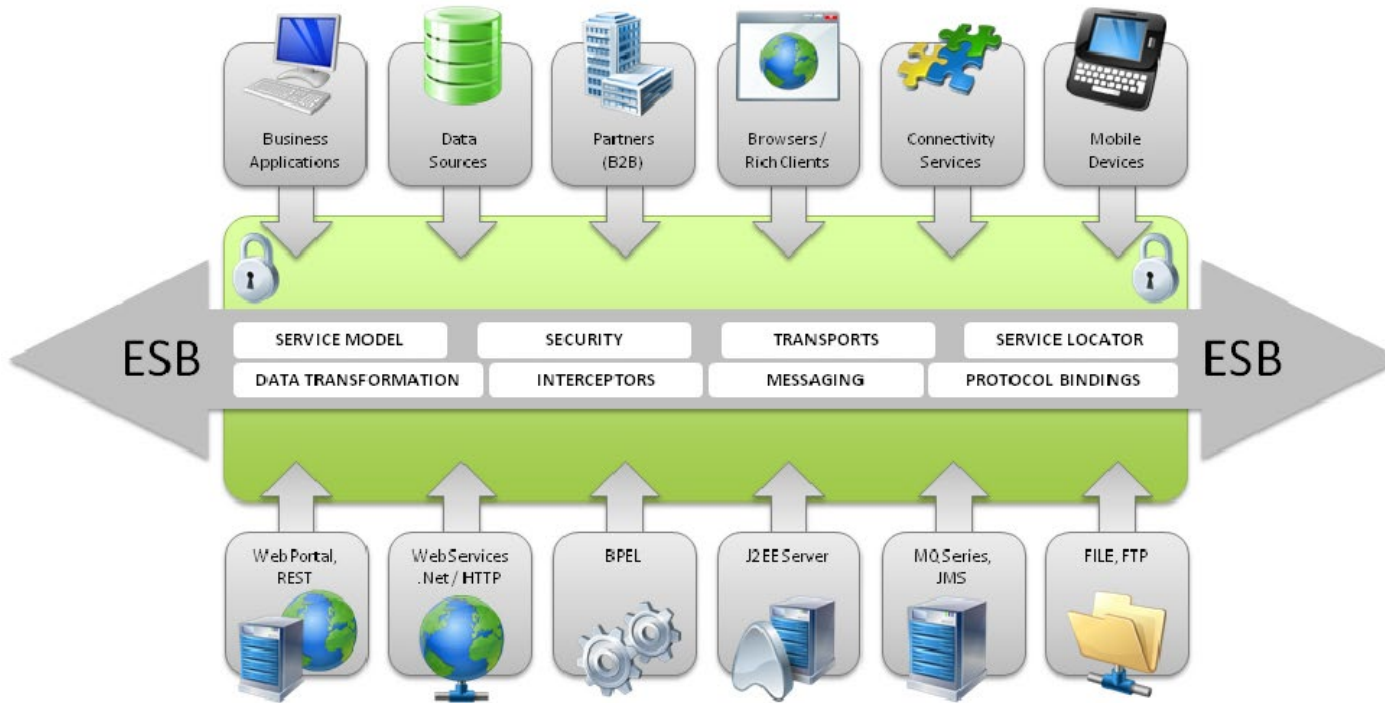
SOA (service-oriented architecture)

- Service-oriented architecture (SOA) is a software development model that allows services to communicate across different platforms and languages to form applications.
- In SOA, a service is a self-contained unit of software designed to complete a specific task.
- Service-oriented architecture allows various services to communicate using a **loose coupling** system



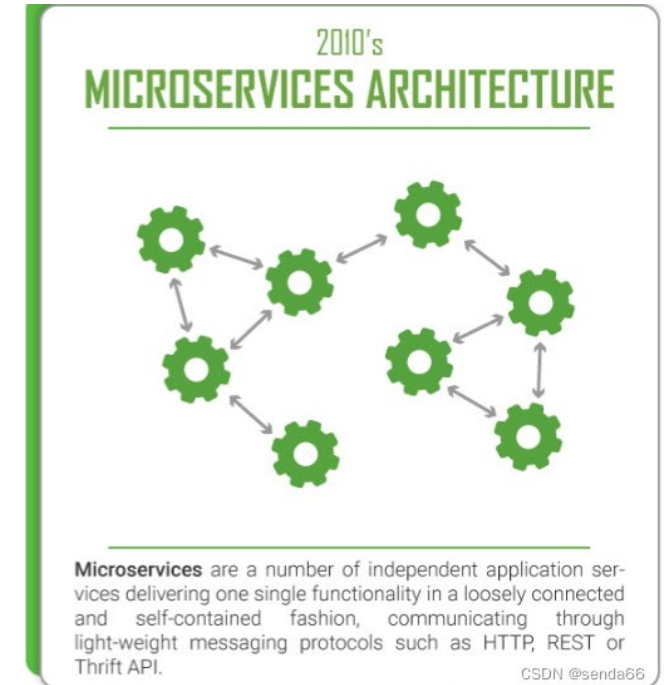
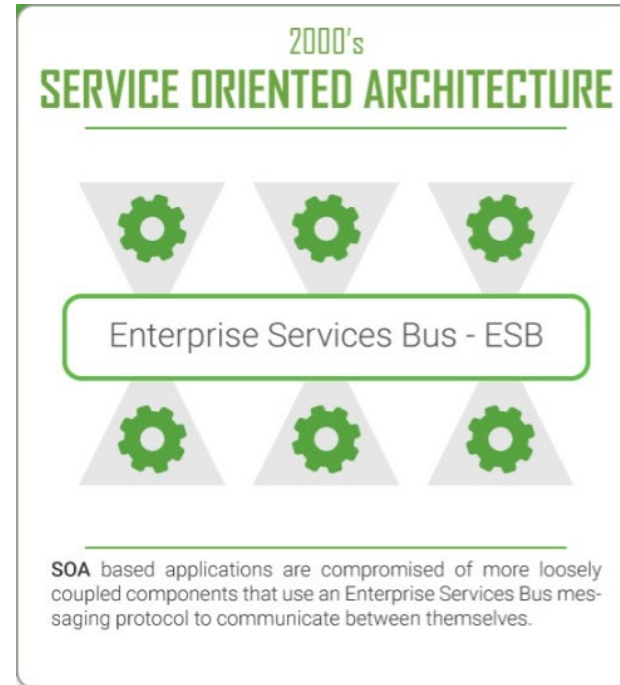
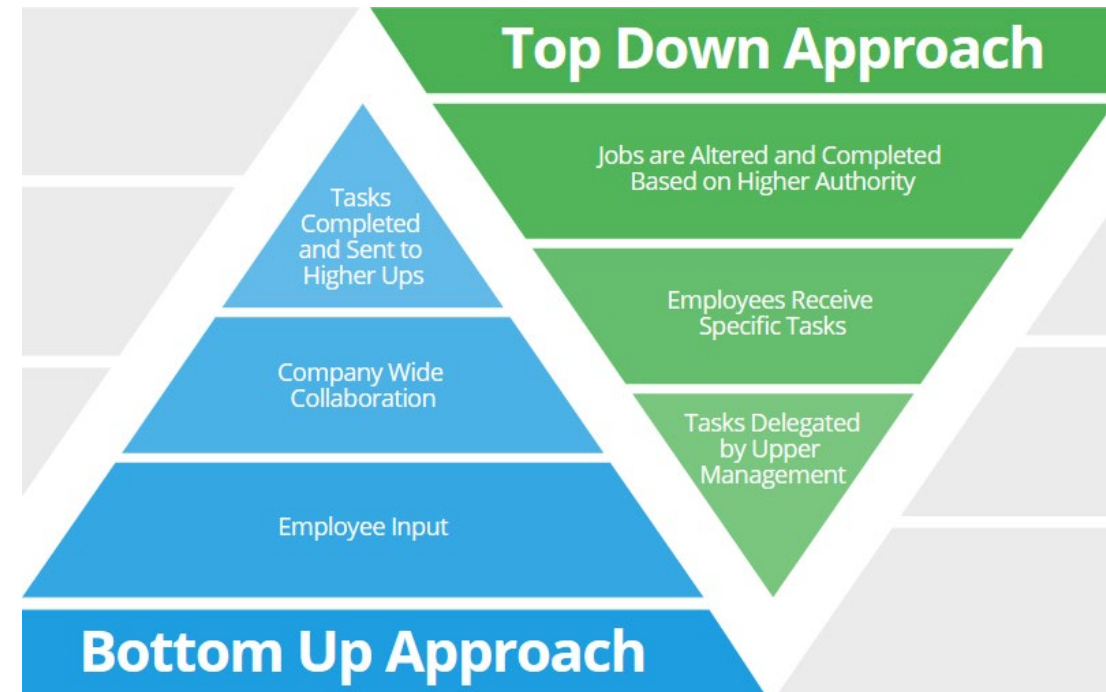
<https://www.techtarget.com/searchapparchitecture/definition/service-oriented-architecture-SOA>
<http://www.taodudu.cc/news/show-4892335.html>

SOA + ESB



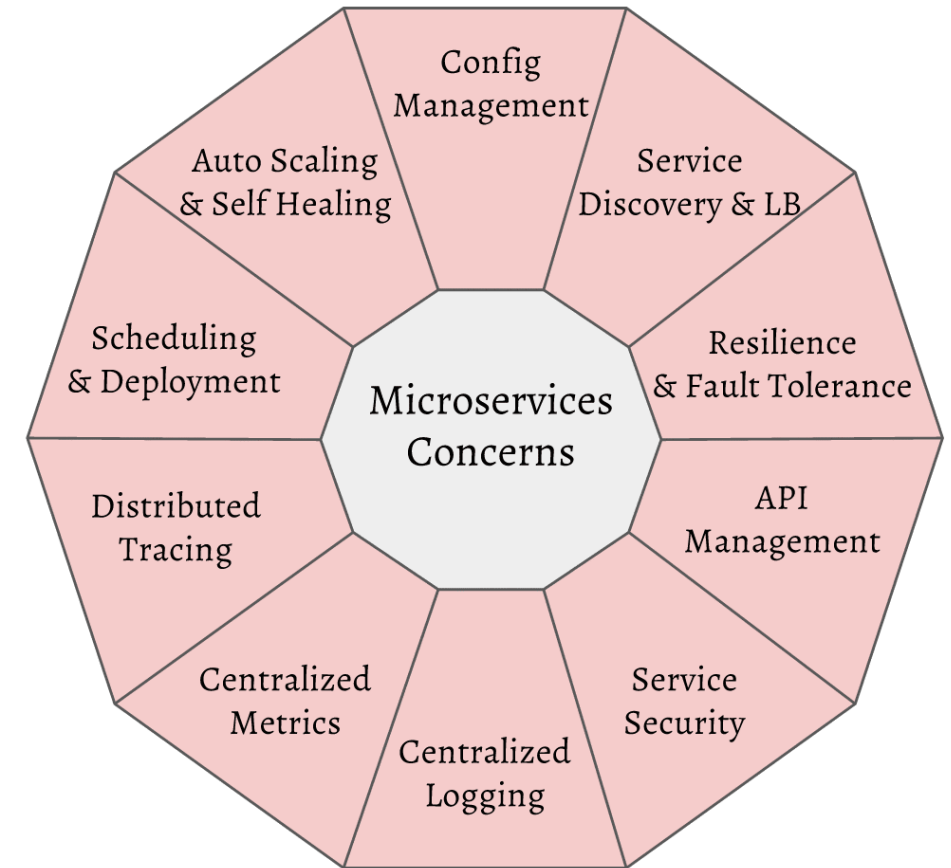
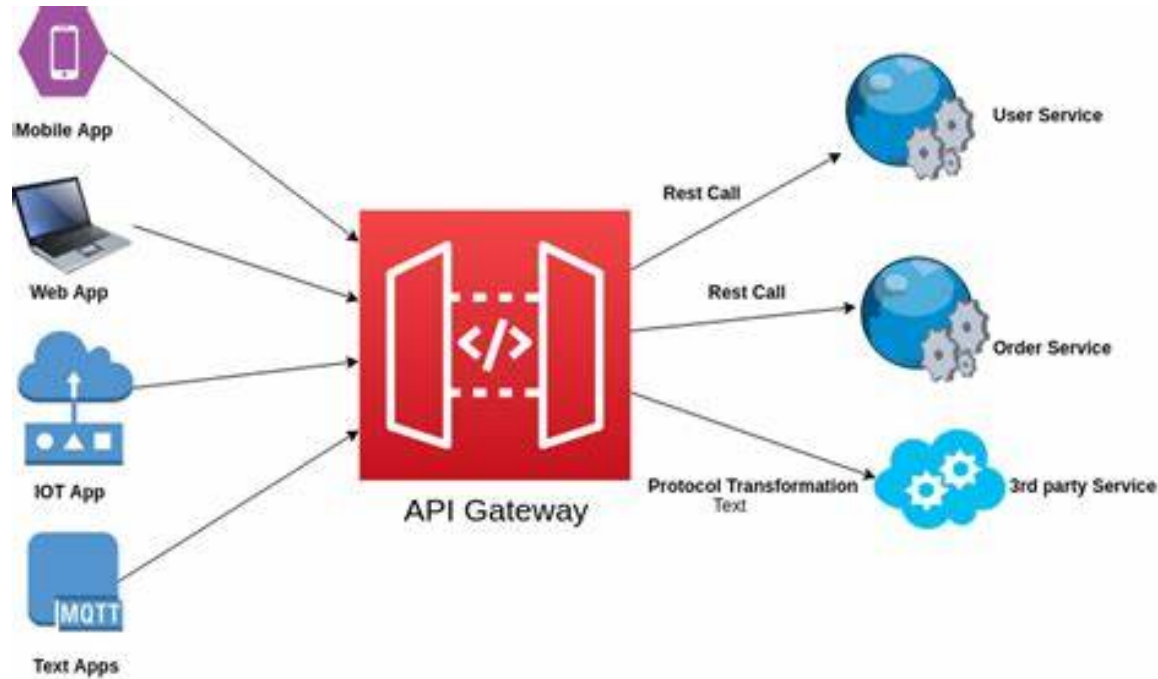
- SOA is an integration architectural style and an **enterprise-wide concept**
- ESB is an implementation architecture. It is a set of rules and principles for integrating numerous applications together over a bus-like infrastructure
- Normally, the enterprise project used ESB solutions with the SOA design

ESB vs Microservice



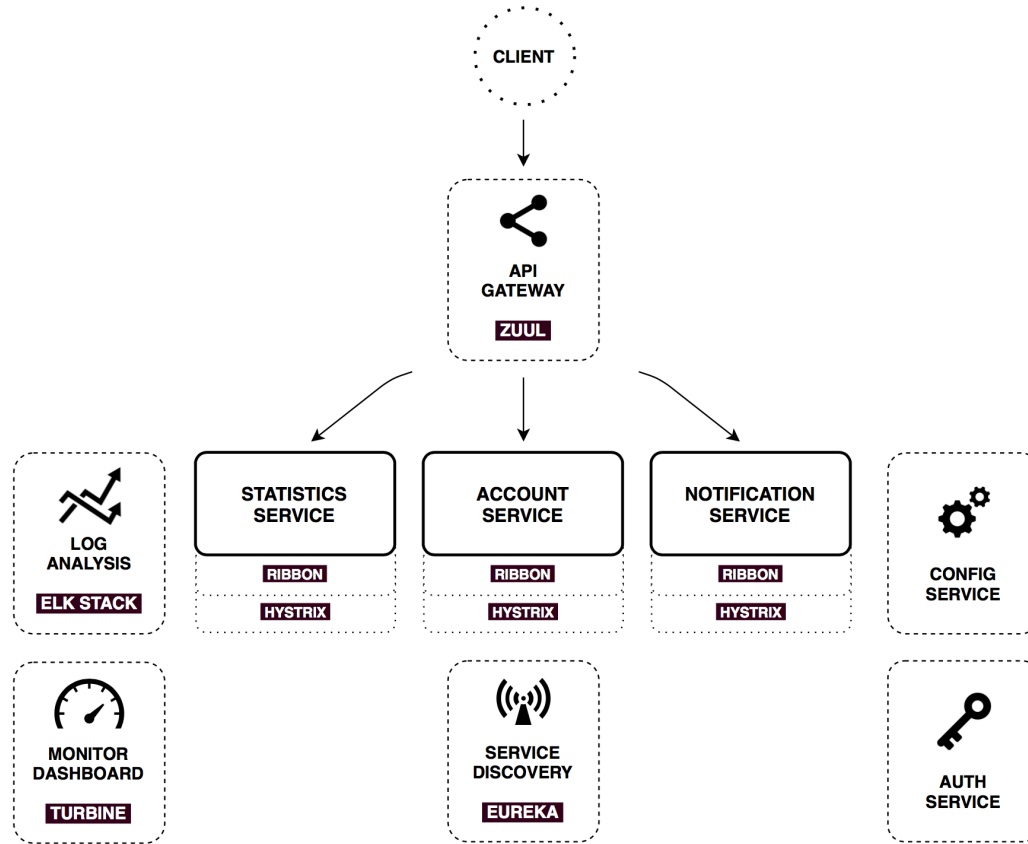
- Normally, ESB is top-down design
- Another approach of SOA bottom-up, an typical one is Microservice

MSA - Microservice Architecture



- Using API gateway to replace **ESB**
- Build the necessary fundamental functions also as **services**

Microservice by Spring cloud



- Spring cloud is from Netflix and be famous "**Spring**" with mature community
- Build-in **services** for fundamental functions
- **Spring boot** is a new way to pack each service with a web server

https://www.sohu.com/a/206889245_115128

<https://dzone.com/articles/microservice-architecture-with-spring-cloud-and-do>

Spring cloud – Service Registry and Discovery



HOME

LAST 1000 SINCE STARTUP

System Status

Environment	test
Data center	default

Current time	2016-04-05T08:25:27 +0000
Uptime	00:03
Lease expiration enabled	false
Renews threshold	10
Renews (last min)	6

DS Replicas

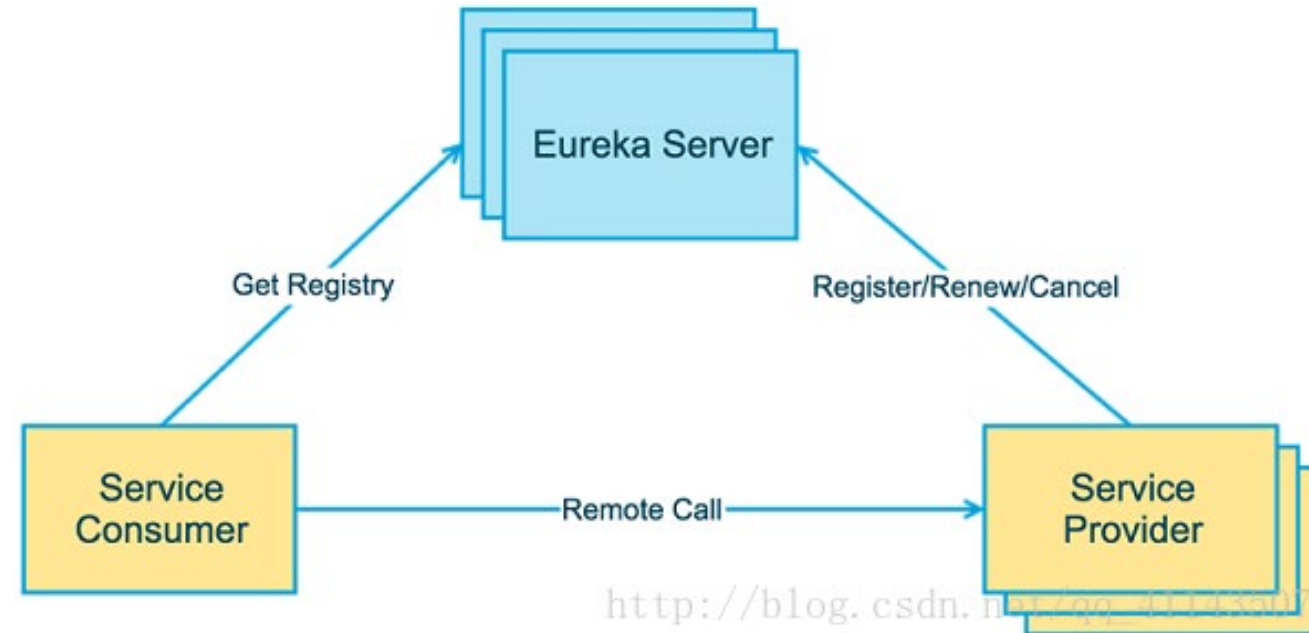
registry

Instances currently registered with Eureka

Application	AMIs	Availability Zones	Status
ACCOUNT-SERVICE	n/a (1)	(1)	UP (1) - 294720aab0a9:account-service:6000
AUTH-SERVICE	n/a (1)	(1)	UP (1) - 8feb1f6a04a2:auth-service:5000
GATEWAY	n/a (1)	(1)	UP (1) - 4dbf6ede836c:gateway:4000
NOTIFICATION-SERVICE	n/a (1)	(1)	UP (1) - 4e91cd36b09b:notification-service:8000
STATISTICS-SERVICE	n/a (1)	(1)	UP (1) - 070cfc12349a:statistics-service:7000

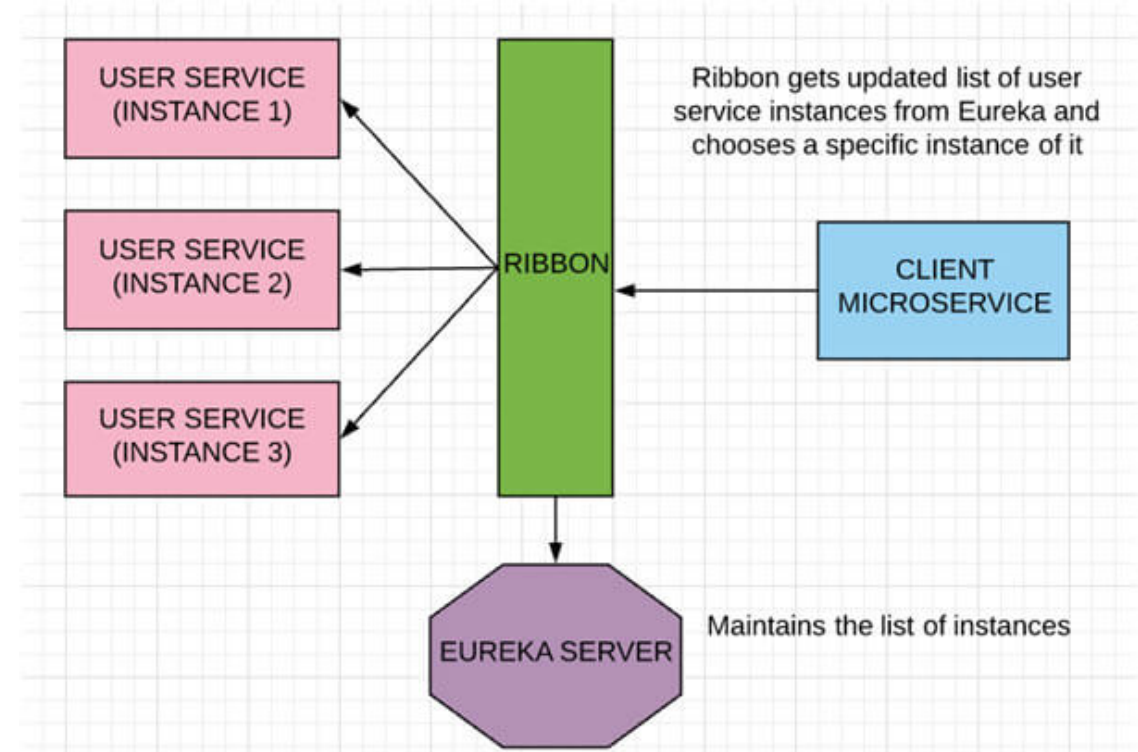
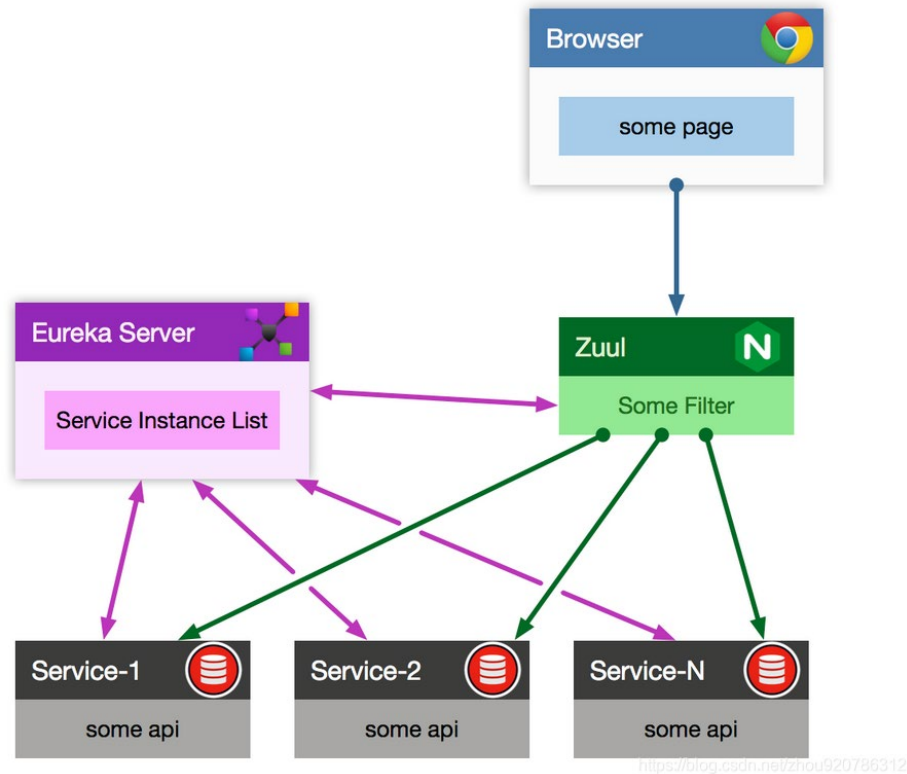
General Info

Name	Value
total-avail-memory	735mb
environment	test
num-of-cpus	4
current-memory-usage	286mb (38%)
server-uptime	00:03



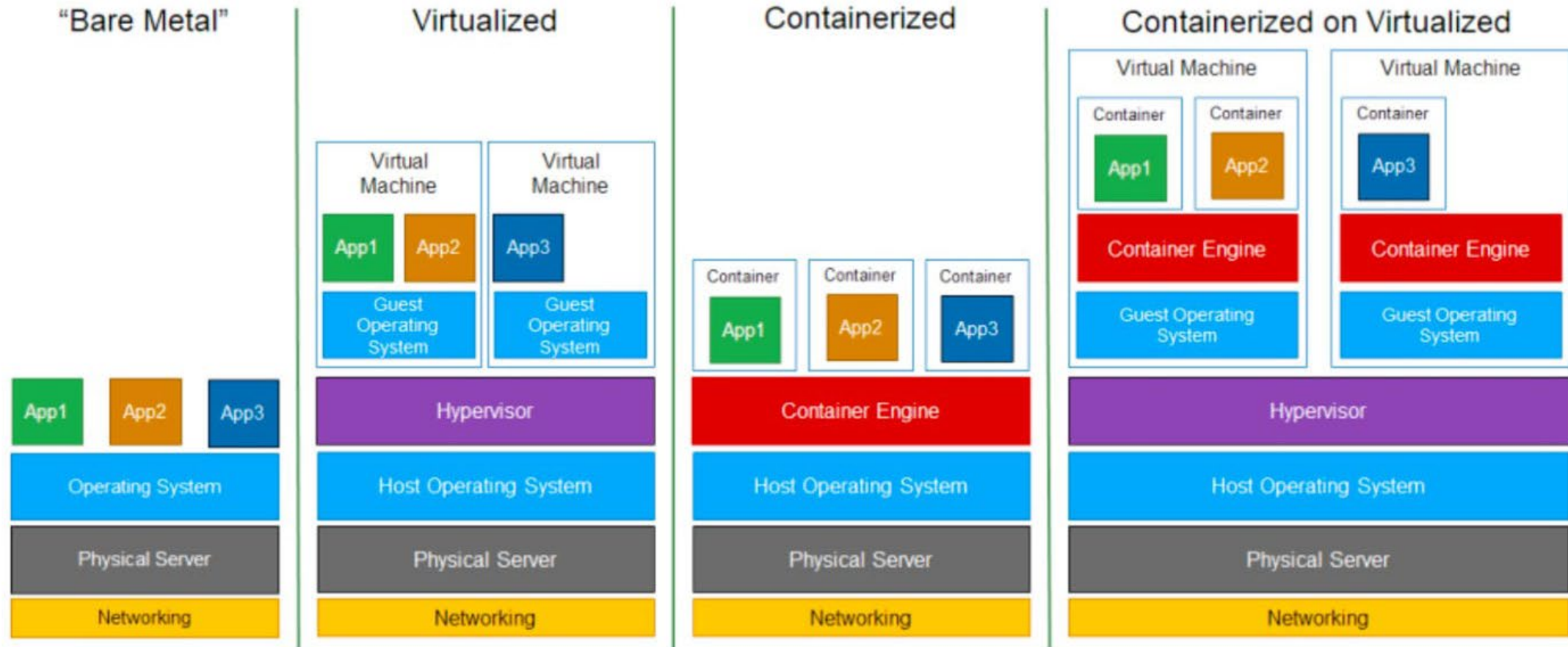
- Dynamic register the providers in Eureka service
- Call the services through Zuul or Feign

Spring cloud – Zuul & Ribbon



- Not all services are open to external
- Zuul worked as a gateway to filter the external requests
- Ribbon is responsible for load balance
- Zuul and Ribbon are build-in microservices also.

Cloud computing



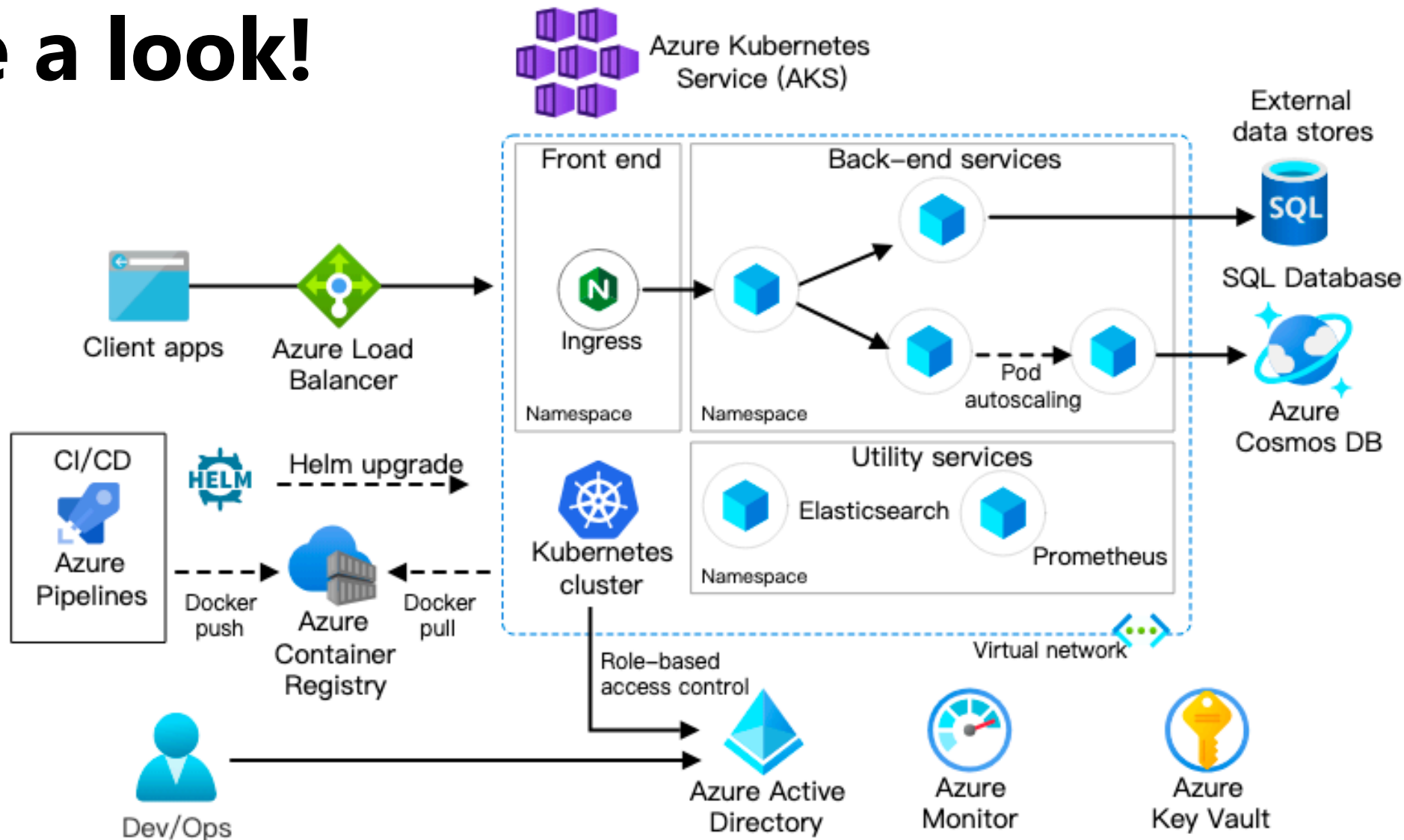
- De-couple the application with hardware
- Hardware are resources pools

Spring cloud vs. Kubernetes

DevOps Experience	Kubernetes	Capability	Spring Cloud with Kubernetes
Auto Scaling & Self Healing		DevOps Experience	Self service, multi-environment capabilities
Resilience & Fault Tolerance		Auto Scaling & Self Healing	Pod/Cluster Autoscaler, HealthIndicator, Scheduler
Distributed Tracing		Resilience & Fault Tolerance	HealthIndicator, Hystrix, HealthCheck, Process Check
Centralized Metrics		Distributed Tracing	Zipkin
Centralized Logging		Centralized Metrics	Heapster, Prometheus, Grafana
API Gateway		Centralized Logging	EFK
Job Management		Job Management	Spring Batch, Scheduled Job
Singleton Application		Load Balancing	Ribbon, Service
Load Balancing	Spring Cloud	Service Discovery	Service
Service Discovery		Configuration Management	Externalized Configurations, ConfigMap, Secret
Configuration Management		Service Logic	Apache Camel, Spring Framework
Application Packaging		Application Packaging	Spring Boot maven plugin
Deployment & Scheduling		Deployment & Scheduling	Deployment strategy, A/B, Canary, Scheduler strategy
Process Isolation		Process Isolation	Docker, Pods
Environment Management		Environment Management	Namespaces, Authorizations
Resource Management		Resource Management	CPU and memory limits, Namespace resource quotas
Operating System		IaaS	GCE, Azure, CenturyLink, VMware, Openstack
Virtualization	IaaS		
Hardware, Storage, Networking			

- Different frameworks may supply the duplicates functions.
- It is a challenge job to choose the RIGHT combinations.

Have a look!

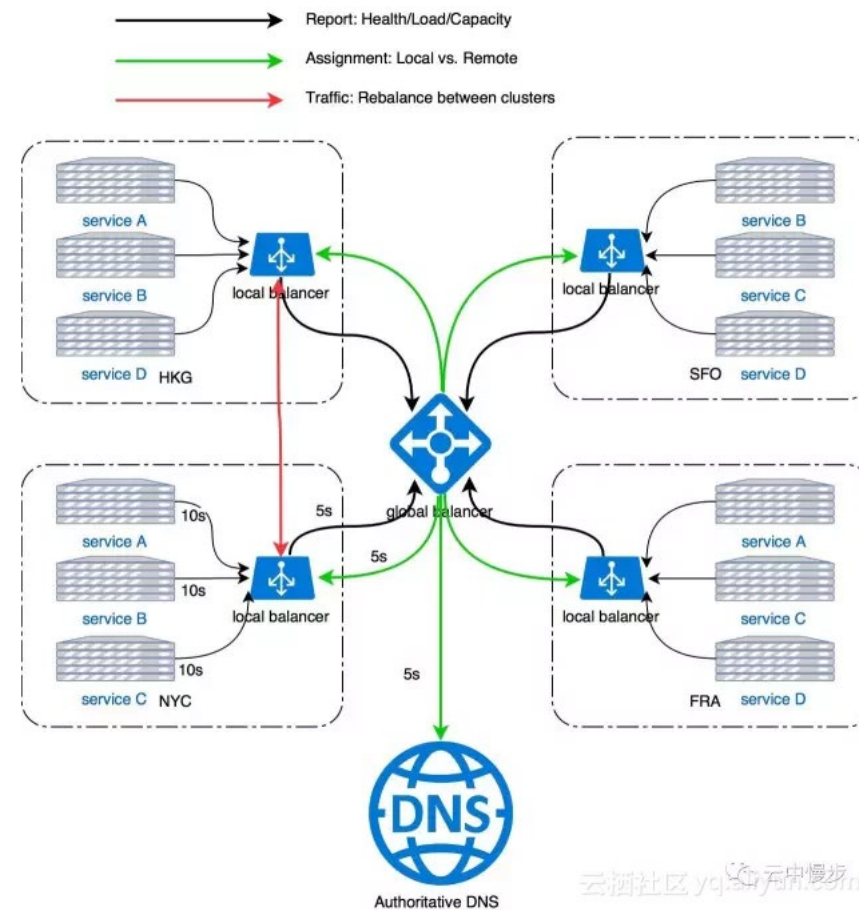


- Many of them are not covered in this module.
- There is no best technologies, choosing the **SUITABLE** one!

DNS balance

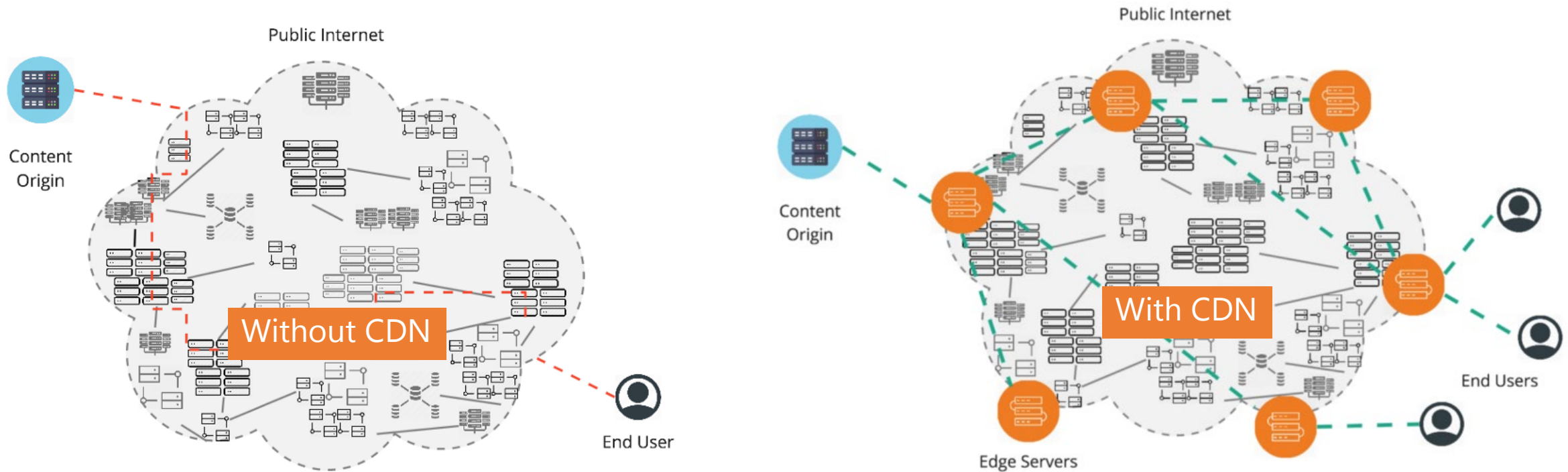


- Split one site to many sites
- Response different IP (corresponding to different site) address to different clients
- Widely used for global applications since long distance net means long delay
- Serve the costumer with the local/nearest site



<https://developer.aliyun.com/article/72614>

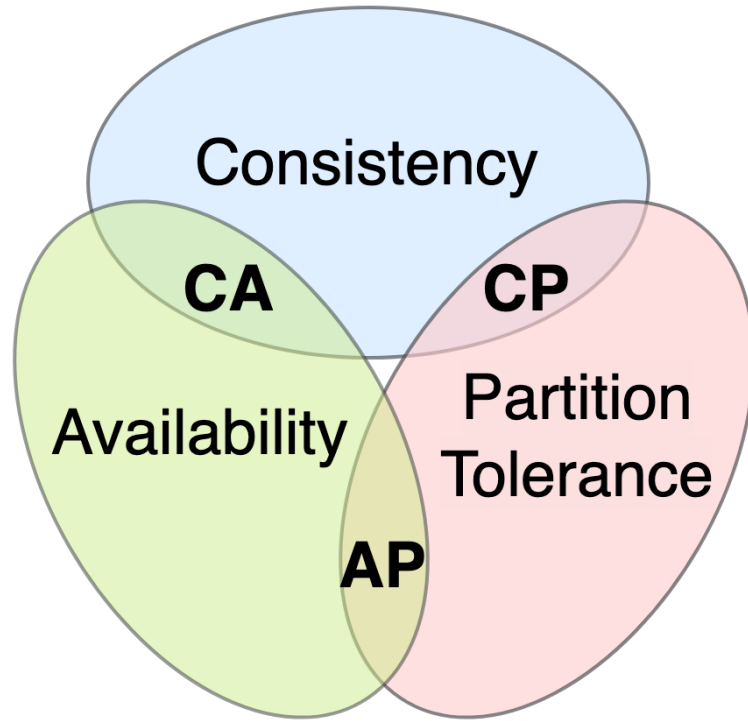
CDN – Content Delivery Network



- CDN means content delivery network, mainly for static resources
- Akamai is the leading company in CDN service
- CDN means you need to have server resources everywhere

CAP theorem

In theoretical computer science, the CAP theorem, also named Brewer's theorem after computer scientist **Eric Brewer**, states that any **distributed** data store can provide only two of the following three guarantees:



Consistency

Every read receives the most recent write or an error.

Availability

Every request receives a (non-error) response, without the guarantee that it contains the most recent write.

Partition tolerance

The system continues to operate despite an arbitrary number of messages being dropped (or delayed) by the network between nodes.